

Who Loses and Why? Market Microstructure of Perpetual Futures

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Perpetual futures trade largely outside reporting infrastructure, so who loses is unknown. On Hyperliquid, a decentralized venue with a complete on-chain record, we classify wallets by how they trade; front-end traders, whom we interpret as predominantly retail, lose \$1.11 billion over fifteen months. Fees account for \$725 million of that loss, net forced exits \$226 million, and voluntary trading \$154 million. Those fees accrue to the venue, not the traders on the other side. Front-end flow follows recent returns, and the post-trade price impact of their voluntary taker trades decays with horizon, reverting past zero after loss-realizing closes.

Keywords: perpetual futures, cryptocurrency derivatives, decentralized exchanges, on-chain data, retail trading, market microstructure, execution costs, liquidations.

JEL Classification: G12, G14, G15, G40, G51.

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1 Introduction

Perpetual futures are futures contracts with no expiration date.¹ Because the contract never expires, a trader can maintain exposure for any horizon without rolling an expiring position into its successor, and trading in each underlying concentrates in a single contract instead of fragmenting across maturities (Ackerer, Hugonnier, and Jermann, 2025; He et al., 2025).² In place of the convergence to spot that a conventional future achieves only at expiration, a funding rate, a periodic payment between long and short positions, ties the contract price to the spot price continuously (Gornall, Rinaldi, and Xiao, 2025). These features have made perpetual futures the dominant instrument in cryptocurrency trading: reported 2025 notional volume reached \$61.7 trillion, more than three times spot volume.³ The Commodity Futures Trading Commission (CFTC) attributes this popularity to the same design, citing “a reduced need for ‘rolling’ a contract from one expiration to another.”⁴

Nor is the contract any longer confined to offshore crypto venues. Cboe, one of the largest traditional exchange operators, has told the CFTC that perpetual futures “may significantly decrease trading costs by eliminating the need to roll expiries.”⁵ In September 2025 the chairs of the Securities and Exchange Commission and the CFTC jointly stated that the two agencies could take steps to “onshore perpetual contracts that meet investor and customer-protection standards” and thereby “capture economic activity now flowing exclusively to foreign platforms.”⁶ In May 2026 the CFTC issued a policy statement on perpetual contracts and permitted the first listing of a bitcoin perpetual as a futures contract on a regulated United States exchange.⁷ The European Union’s markets regulator has stated that perpetual contracts offered to retail clients are likely to fall within its product-intervention measures, which impose leverage limits, mandatory margin close-out, and negative balance protection.⁸ Protection instruments of this kind presume knowledge of how the product is actually traded: who participates, how often their positions are forcibly closed, what they pay, and who ultimately bears the losses.

For perpetual futures, that knowledge is scarce, because the market being brought onshore developed offshore. In the CFTC’s words, perpetual-futures trading has “largely developed outside

¹BitMEX announced its XBTUSD perpetual swap in May 2016. See BitMEX, May 13, 2016.

²Practitioners emphasize the same feature: perpetual futures “enable continuous exposure without the need to repeatedly roll expiring contracts.” See Don Wilson (DRW), X, July 28, 2026.

³Reuters, citing CryptoQuant, reports \$61.7 trillion in 2025 perpetual-futures volume and \$18.6 trillion in spot volume. See Reuters, April 22, 2026.

⁴CFTC, Request for Comment on the Trading and Clearing of “Perpetual” Style Derivatives, Release 9069-25, April 21, 2025. See CFTC Release 9069-25.

⁵Cboe Global Markets, comment letter to the CFTC on Release 9069-25, May 21, 2025.

⁶Paul S. Atkins and Caroline D. Pham, Joint Statement from the Chairman of the SEC and Acting Chairman of the CFTC, September 5, 2025.

⁷CFTC, Policy Statement Concerning the Listing of Perpetual Contracts, 91 Fed. Reg. 33160 (June 3, 2026), adopted alongside an order permitting a regulated exchange to list a bitcoin-referencing perpetual contract. See 91 FR 33160.

⁸ESMA, Public Statement ESMA35-243228190-8024, February 24, 2026.

of the United States,” on venues that sit outside the reporting infrastructure of regulated markets: there is no consolidated audit trail, no position reporting, and accounts are pseudonymous.⁹ Regulators are thus designing protection for trading behavior they cannot observe. Who trades perpetual futures, who gains, who loses, and through which trading mechanisms are, for these markets, open questions.

What the public record shows instead is anecdotes. Reported forced liquidations across crypto derivatives reached roughly \$150 billion in 2025, including more than \$19 billion on October 10 alone.¹⁰ Individual accounts can experience similarly large reversals: one widely reported account moved from approximately \$100 million in profit to a \$17.5 million loss within weeks.¹¹ These anecdotes suggest large transfers of wealth, concentrated in forced exits and borne in large part by retail participants, the very population the protection instruments exist for. What they cannot establish is the magnitudes, who exactly bears them, or the mechanisms through which they occur.

We answer these questions for one venue whose complete record is public: Hyperliquid, a leading decentralized perpetual-futures exchange. Our sample covers \$2.95 trillion of trade volume from March 27, 2025 through June 30, 2026 across the ten perpetual contracts with the highest trading volume on Hyperliquid in the second half of 2025. It is the largest decentralized perpetual-futures venue and, by open interest, the second largest overall, holding \$9.3 billion against Binance’s \$22.1 billion (CoinGecko, 2026a). Hyperliquid is one venue, not the market: its scale makes the evidence economically significant, but the results describe trading on this venue, and we read them as insight into perpetual-futures markets rather than a census of them.

Hyperliquid runs its matching engine on a public blockchain, so its complete record is observable: orders, cancellations, trades, and forced liquidations are recorded at millisecond resolution, together with each account’s position and realized profit and loss. Observing everything, however, identifies no one. Even in traditional markets, where brokers and regulators hold account records, researchers must infer which trades come from retail investors (Boehmer et al., 2021), and the institutional structure that inference relies on has no counterpart here. Accounts are pseudonymous wallets: the record shows everything a wallet does and nothing about who it is.

Our first contribution is therefore a new way to identify retail-like participants from trading behavior alone. The classification takes advantage of two features of the venue’s records that are not obvious: an order submitted through Hyperliquid’s own web or mobile interface is recorded as its own order type, and a resting order that can only add liquidity carries a dedicated designation. We classify wallets by how they trade. *Algorithmic market makers* continuously post, update, and cancel resting limit orders on both sides of the book, mostly under the add-liquidity-only designation.

⁹91 Fed. Reg. at 33160.

¹⁰See CryptoSlate, reporting CoinGlass data, and CoinDesk.

¹¹CoinDesk, June 2, 2025.

Front-end traders place orders through the venue’s interface, as a person trading manually does; we interpret them as predominantly retail. *Algorithmic takers* are automated accounts that mostly take liquidity. A small *Ambiguous* group (1.7% of volume) fits none of these signatures and is excluded from the outcome analysis.

Our second contribution turns this classification into an accounting of who gains and who loses. We compute and decompose each group’s profit and loss trade by trade, net of funding and fees. Over our fifteen-month sample front-end traders lose \$1.11 billion, algorithmic market makers gain \$212 million, and algorithmic takers approximately break even. Front-end traders finance the rest of the venue, and the traders on the other side are not the main beneficiary: the largest share of the shortfall goes to fees, more than three times what the algorithmic market makers gain, and fees accrue to the exchange itself.

Why do front-end traders lose more than a billion dollars in a market where every wallet trades against the same order book under the same posted fee schedule? The answer begins with the decomposition: \$725 million of the shortfall is fees, \$226 million is net forced exits, and only \$154 million comes from voluntary trading.

Our third contribution isolates the mechanism behind each loss channel. We examine them in turn: voluntary trading, net forced exits, and fees.

In voluntary trading, front-end traders pay 2.43 bps of execution cost per dollar of taker volume, against 0.70 bps for algorithmic market makers and 1.70 bps for algorithmic takers. The gap is not in the half-spread — the cost of reaching the first execution price — which the three groups pay at close to the same rate. It is in slippage, the extra cost of the depth an order consumes beyond that first price. The natural suspect is order size: slippage rises sharply with the size of the aggressive order, and front-end volume is more concentrated in large aggressive orders than that of the other two groups. Yet even conditional on order size, algorithmic market makers pay less. The remaining difference is depth sensitivity: front-end traders barely scale order size to opposing depth, while algorithmic market makers do.

The second component of voluntary trading is post-trade price impact: the change in the midprice over a fixed horizon, signed by the direction of the trade. Initially, all three groups gain, and after one minute their price impacts are nearly identical: 2.44 bps for front-end traders, 2.55 bps for algorithmic market makers, and 2.42 bps for algorithmic takers. By two hours, however, the paths diverge: front-end impact falls to 1.23 bps, while market-maker impact holds at 2.62 bps and taker impact rises to 3.09 bps.¹² Why does the front-end path decay over those two hours while both algorithmic paths rise?

The answer lies in the purpose of the trade. We partition voluntary taker trades into openings, which increase a position, gain-realizing closes, which reduce a position at a profit, and loss-

¹²This horizon comparison excludes October 10, 2025, the day of the marketwide liquidation cascade.

realizing closes, which reduce a position at a loss. Loss-realizing closes are the worst for every group: they are urgent trades, a loss being cut is cut quickly, and price impact declines between one minute and two hours after them for all three groups. For front-end traders the decline is by far the steepest: their post-trade price movement reverts past zero, from +2.61 bps one minute after the trade to −0.71 bps two hours after it. The flow regressions show who is on each side: front-end loss-realizing flow follows recent returns, selling into declines or buying into rises to cut a loss, that is, momentum, while opening and gain-realizing flow from both algorithmic groups is contrarian, so the algorithmic groups stand on the other side. Declines are followed by more loss-realizing selling, and algorithmic-market-maker flow moves against it in the minutes that follow.

Openings add a second gap, this one across groups. Front-end opening flow also follows recent returns, the only momentum openings among the three groups, while both algorithmic groups open against the market's recent direction. And front-end openings are the only ones whose price impact declines with horizon, from 2.20 bps at one minute to 1.64 bps at two hours, while algorithmic market makers rise from 2.79 to 3.95 bps and algorithmic takers from 2.38 to 3.57 bps.

Gain-realizing closes hold up for all three groups, with contrarian flow for all three. Front-end voluntary losses therefore come from both margins: loss-cutting trades whose price impact decays for every group but decays most steeply for them, and opening trades where they alone chase recent returns.

A loss-realizing close is a voluntary exit from a losing position, and a liquidation is the forced version of the same exit, made once account equity falls below the required margin, with the protocol rather than the trader picking the moment. On ordinary days it is front-end traders who bear the liquidations: outside October 10 they account for 97.6% of the liquidation volume of the three groups, and only 9.3% of their liquidation volume falls on that single day, while 79.7% of algorithmic market-maker liquidation volume does. They are also very costly, and it is front-end traders who pay for them: liquidations pay an average half-spread of 15.16 bps, and, outside October 10, the price keeps moving against them, from 16.0 bps five minutes after the trade to 28.5 bps two hours after it. Auto-deleveraging, which closes profitable positions on the other side of the market when a liquidation cannot clear against the book, returns part of that loss.

Fees, the last channel, are set by the schedule itself rather than by any price a trader receives. The maker–taker schedule charges the taker more than the maker at every tier and rebates the highest-volume makers, and the tier a wallet lands in depends on its recent trading volume and on how much HYPE it stakes.¹³ Front-end traders sit on the costly side of both dimensions: they take liquidity on 85% of their volume, against 20% for algorithmic market makers, and most of their volume comes from wallets in the base tier with no stake, while algorithmic market-maker volume concentrates in the lower-fee tiers. Role and tier together produce \$725 million in fees

¹³Hyperliquid, “Trading Fees,” Hyperliquid Docs (accessed August 12, 2026).

from front-end traders, against \$50 million from algorithmic market makers and \$55 million from algorithmic takers. Of the fees the three groups pay, the venue collects 87% from front-end traders.

The rest of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 introduces the institutional setting. Section 4 describes the data. Section 5 defines the trading-cost decomposition, the trade-purpose partition, and the wallet classification. Section 6 presents the empirical results. Section 7 concludes. The Appendix collects descriptive tables, robustness checks, and supplementary regression panels.

2 Related Literature

Our paper connects to four strands of literature.

The first strand studies perpetual futures. On pricing, the conceptual origin lies in the proposal of cash-settled perpetual claims on illiquid asset indices (Shiller, 1993). Akerer, Hugonnier, and Jermann (2025) derive no-arbitrage prices for perpetual contracts under the funding-payment mechanism, and He et al. (2025) give a theoretical and empirical account of the funding rate. On clearing and margin risk, auto-deleveraging has been modeled as the loss-socialization mechanism that operates once ordinary liquidation fails (Chitra, 2026; Campbell et al., 2026). On microstructure, Ruan and Streltsov (2025) show that perpetual listings affect spot-market liquidity, spreads, and informed trading around funding cycles, and Gornall, Rinaldi, and Xiao (2025) show that perpetuals reduce extreme price dislocations relative to quarterly futures. This work models the contract’s pricing, clearing, and microstructure; we add its missing empirical foundation: who trades on these venues, who gains and who loses, and which channels carry the transfer, namely fees, net forced exits, and voluntary trading.

A second strand studies decentralized-exchange microstructure. It has mostly studied spot automated market makers: trading and market quality (Lehar and Parlour, 2025; Capponi, Jia, and Yu, 2024; Barbon and Rinaldo, 2026), liquidity provision and fee design (Aoyagi, 2020; Hasbrouck, Rivera, and Saleh, 2025a,b; Capponi and Jia, 2025; Lehar, Parlour, and Zoican, 2024), and frictions specific to that design, such as just-in-time liquidity, loss-versus-rebalancing, sandwich risk, validator rents, and maximal extractable value (Park, 2023; Capponi, Jia, and Zhu, 2024; Milionis et al., 2024; Malinova and Park, 2024; Cartea, Drissi, and Monga, 2024; Aquilina et al., 2024; Daian et al., 2020). These papers establish blockchains as a useful laboratory for market microstructure, because trading rules and the wallet-level events they generate are publicly observable. Wallet-level evidence on decentralized perpetual-futures trading, by contrast, remains scarce. We use the same transparency on a different economic object, the on-chain perpetual-futures limit order book, and treat it as a laboratory for the much larger off-chain perpetual market, where such records are rarely available.

A third strand studies trading, price formation, and market integrity in cryptocurrency markets. Makarov and Schoar (2020) document recurrent price deviations across exchanges and show that arbitrage integrates these markets only slowly. Hu and Zhang (2026) show that arbitrageurs organize fragmented exchanges into core–periphery liquidity networks. Related work links liquidity and price-discovery measures to subsequent price dynamics (Easley et al., 2026). A market-integrity branch documents fake volume, stablecoin-related price support, and pump-and-dump schemes (Cong et al., 2023; Griffin and Shams, 2020; Li, Shin, and Wang, 2025). These papers also raise a measurement problem, because much of the evidence rests on exchange-reported records from venues whose reporting quality is in question. Our ledger is recorded by the protocol rather than reported by the exchange, which lets realized gains and losses be allocated across participant types.

A fourth strand studies retail trading. Account-level and market-wide studies show that individual investors trade actively and underperform after trading costs (Odean, 1999; Barber and Odean, 2000; Barber et al., 2009). Later work shows that retail order flow is heterogeneous, since retail trades can supply liquidity or carry information, and the inference depends on order type and measurement method (Kaniel, Saar, and Titman, 2008; Kelley and Tetlock, 2013; Linnainmaa, 2010; Barber et al., 2024; Barber, Lin, and Odean, 2024; Cannon and Mohrschladt, 2026). A related execution-quality literature studies how routing, venue design, intermediary incentives, and order size shape the costs retail investors pay (Battalio, Corwin, and Jennings, 2016; Schwarz et al., 2025; Dyhrberg, Shkilko, and Werner, 2025). Recent work extends the analysis to speculative and leveraged assets, including options, futures, and cryptocurrencies, and to digital interfaces that shape attention, risk taking, and trading intensity (Bryzgalova, Pavlova, and Sikorskaya, 2023; Subrahmanyam et al., 2024; Heimer and Simsek, 2019; Kogan et al., 2024; Chapkovski, Khapko, and Zoican, 2026; Grant, Hobson, and Sinha, 2024). Because exchanges rarely label the account behind a trade, this literature identifies retail flow from institutional structure: Boehmer et al. (2021) identify a broad swath of marketable retail orders from the subpenny price improvement that wholesalers give them. Our market has none of that structure: no wholesaler routes its orders, and no broker labels its accounts.

Relative to these literatures, we make three main contributions. First, we provide a classification that identifies retail-like participants from on-chain order and fill signatures alone, without broker labels, account identifiers, or routing information. Second, we provide a group-level accounting of who gains and who loses in a perpetual-futures market, measured trade by trade and net of funding and fees. Third, we provide a mechanism account of the three channels through which the transfer runs, namely fees, net forced exits, and voluntary trading, and we measure how much each carries. Designing investor-protection rules for perpetual markets requires evidence of exactly this kind.

3 Institutional Background

3.1 Perpetual Futures

A perpetual futures contract lets a trader take a long or short position on an underlying asset without owning it. The trader posts margin collateral rather than the full position value, so the position is leveraged. Unlike standard futures, the contract does not expire. Instead, its price is tied to the underlying spot market through periodic funding payments between longs and shorts: when the perpetual trades above spot, longs pay shorts, and vice versa; see Ackerer, Hugonnier, and Jermann (2025) for a detailed treatment of the contract and its pricing under this mechanism. Funding settles every hour on Hyperliquid and at intervals ranging from one to eight hours on major centralized venues, with a baseline rate of one basis point per eight hours and caps that allow up to four percent per hour in stressed markets.¹⁴ Perpetual futures are one of the dominant traded products in cryptocurrency markets: aggregate perpetual volume runs at roughly three to four times the daily spot volume in the same underlyings (CoinGecko, 2026b).

3.2 Decentralized Exchanges

Perpetual futures trade on exchanges that are typically offshore and clear without a clearinghouse. The exchange itself maintains margin: it marks account equity to market continuously and liquidates positions whose equity falls below the maintenance requirement. When a liquidation cannot cover the shortfall, the exchange absorbs it through auto-deleveraging or insurance mechanisms; see Campbell et al. (2026) and Chitra (2026) for detailed treatments of the auto-deleveraging mechanism and the liquidation process.

Trading of perpetual futures occurs on both centralized and decentralized exchanges. A centralized exchange runs its order book and matching engine, computes account balances and risk, and settles trades in a private database controlled by the operator; a decentralized exchange moves some of these functions onto a public blockchain, where trades, settlement, or the matching rules themselves can be inspected directly, though how much moves on chain varies from venue to venue. Centralized venues dominate perpetual volume, but they do not publicly expose persistent trader accounts; decentralized venues are smaller, settle on public blockchains, and reveal wallet-level activity directly, and their share of perpetual open interest has grown rapidly, from about 3.6% in January 2025 to roughly 16% in July 2026 (CoinGecko, 2026a). In spot trading the dominant decentralized design is the automated market maker, in which traders trade against pooled liquidity at protocol-set prices; perpetual futures instead typically trade on order-book venues, where traders

¹⁴Hyperliquid, “Funding,” Hyperliquid Docs (accessed August 12, 2026): funding is paid hourly, the interest component is 0.01% per eight hours, and funding is capped at 4% per hour.

post limit orders, marketable orders consume resting depth, and leverage, funding payments, and risk management are layered on the limit-order infrastructure.

Within the decentralized segment, the market is highly concentrated around Hyperliquid, the largest crypto perpetual DEX. As of July 1, 2026, Hyperliquid was the second-largest perpetual-futures venue by open interest across centralized and decentralized exchanges, holding \$9.3 billion against Binance’s \$22.1 billion (CoinGecko, 2026a); its July 2026 perpetual volume equaled about 7.8% of the combined volume of the fourteen largest centralized exchanges (The Block, 2026b). As of August 2026, Hyperliquid averaged roughly \$6.2 billion per day in perpetual volume over the trailing month, had accumulated more than \$5.0 trillion in cumulative volume, and carried about \$10.9 billion in open interest (DefiLlama, 2026). This makes Hyperliquid a natural representative venue for studying perpetual-futures microstructure with wallet-level data.¹⁵

3.3 Hyperliquid

Hyperliquid is a Layer 1 blockchain built for high-throughput trading via an on-chain limit order book. As of Q2 2026, Hyperliquid lists over 200 perpetual futures contracts and a separate spot contract list. Its entire order book and matching engine run on-chain. Validators stake the native HYPE token to produce blocks, and each block contains the ordered sequence of exchange events: order submissions, cancellations, matched trades, liquidations, transfers, and funding-related events. Order latency is approximately 0.2 s at the median and 0.9 s at the 99th percentile for co-located clients, with sustained throughput of approximately 200,000 orders per second.

3.4 Vaults, Liquidations, and the October 10 Event

On Hyperliquid, a liquidation begins with a market order for the full position size, sent by the protocol to the book, which may close the position in whole or in part.¹⁶ If the book does not absorb the order and account equity falls below two-thirds of the maintenance margin, the residual position passes to the liquidator vault, a strategy of the Hyperliquidity Provider (HLP), which is Hyperliquid’s own liquidity-provider vault. The vault takes the residual position onto protocol capital and unwinds it on a time-weighted schedule; the gains and losses of these backstop exits accrue to HLP’s depositors. When an account’s value has already gone negative and the position still cannot be cleared, auto-deleveraging closes opposite-side positions, ranked by unrealized profit

¹⁵Part of Hyperliquid’s most recent growth comes from builder-deployed markets (HIP-3), dominated by tokenized-equity perpetuals, which accounted for roughly one third of its open interest and about half of its trading volume in July 2026 (The Block, 2026a). Our sample ends June 30, 2026 and covers ten native crypto perpetual contracts, so it does not include builder-deployed markets.

¹⁶Hyperliquid, “Liquidations,” Hyperliquid Docs (accessed August 12, 2026). Four Hyperliquid protocol accounts stand behind this process: two HLP vaults and two liquidator engines.

and leverage used, at the previous mark price against the underwater account.¹⁷

On October 10, 2025, a marketwide sell-off in cryptocurrencies produced the largest forced-liquidation day on record, with more than \$19 billion of positions liquidated across venues.¹⁸ On Hyperliquid the cascade liquidated positions in more than six thousand wallets and triggered auto-deleveraging across the venue.¹⁹ The analysis that follows treats this day separately where its inclusion changes a comparison.

4 Data

The Hyperliquid network can be accessed by running public non-validator nodes that continuously stream and replay the full sequence of committed blocks via a gossip protocol, writing approximately 100 GB of data per day. The chain produces state snapshots every 10,000 blocks, capturing all users’ account balances, open positions, leverage settings, accrued funding, and the full limit order book across every listed market. Together, the snapshots and inter-snapshot block streams provide a complete, tick-level reconstruction of all trading activity and state evolution on the exchange, forming the basis of the dataset used in this study.

Our analysis combines wallet-level order and execution records with reconstructed order-book states. Orders and fills are obtained through Allium, which indexes Hyperliquid’s public order and fill records, and top-of-book snapshots are obtained from Tardis. The sample runs from March 27, 2025 through June 30, 2026 and covers the ten perpetual contracts with the highest trading volume in the second half of 2025: BTC, ETH, SOL, HYPE, XRP, DOGE, ENA, FARTCOIN, SUI, and LINK (see Table 1).²⁰ Across the sample, the ten coins account for about \$2.95 trillion of trade notional.

The **orders** table records the lifecycle of each order at millisecond resolution. A single order generates an initial submission row and a later terminal row when the order is filled, canceled, or rejected. Each row indicates the user, timestamp, order id, price, side, order type, and time-in-force. The time-in-force field determines the execution policy. *Alo* denotes add-liquidity-only orders: post-only limit orders that either rest on the book or are rejected if they would execute immediately. *Gtc* denotes good-till-cancelled limit orders, which can rest passively or execute aggressively if marketable at submission. *Ioc* denotes immediate-or-cancel limit orders: they execute immediately against resting liquidity up to their limit price, with any unfilled quantity cancelled. *FrontendMarket*

¹⁷Hyperliquid, “Auto-Deleveraging,” Hyperliquid Docs (accessed August 12, 2026).

¹⁸See CryptoSlate, reporting CoinGlass data.

¹⁹CoinDesk, October 11, 2025.

²⁰The sample starts on March 27, 2025 because the upstream orders feed is complete from this date onward. Earlier records have incomplete order and time-in-force coverage; in particular, *Alo*, which identifies post-only liquidity provision, was not reliably observed.

identifies market orders submitted through Hyperliquid’s native web or mobile interface, while *LiquidationMarket* identifies system-generated market orders used for liquidations.

The **fills** table records executed trades. Each fill appears as two rows, taker and maker, with a common trade id, and relates to the orders table with order id. A single order can match multiple counterparties or walk multiple price levels, so one order id can generate many fill rows. The fills table records the user’s address, pre-trade position, realized closed P&L, fees, liquidation status, and a *direction* field that identifies whether the execution opens, closes, or reverses a long or short position, and separately flags protocol-initiated liquidations and auto-deleveraging events.

The fills and orders tables do not contain the contemporaneous bid and ask quotes needed to measure execution costs. We therefore build an order-book panel by combining the order and fill streams with Tardis book snapshots. The Tardis feed provides periodic observations of the best bid, best ask, and displayed depth at the top 5 levels around every 500 ms. Because these snapshots are not event-aligned with trade timestamps, we use them as synchronization anchors and replay the order and fill streams between snapshots to recover the book state around each trade. For each fill, we recover the strict-prior best bid and ask, b_t and a_t , and define the contemporaneous midprice as $m_t = (b_t + a_t)/2$.²¹ Forward midprices m_{t+h} are taken from the same replayed book panel. The strict-prior convention is necessary because Allium timestamps are recorded in milliseconds and do not preserve the exchange engine’s within-millisecond ordering. Events that share the same millisecond cannot be sequenced from the data alone, so execution-cost calculations use the book state immediately before that millisecond.

5 Empirical Methodology

This section classifies every wallet into a behavioral group and then defines the per-trade objects whose group-level values Section 6 reports.

5.1 Wallet Classification

We classify wallets into three trader groups: algorithmic market makers (MM), front-end traders (Frontend), and algorithmic takers (Algo). In every match one side is the taker, whose arriving order removes resting liquidity, and the other side is the maker, whose resting order is executed against. Most of the classification rests on two features of the order record, which we describe, together with what each reveals about a wallet, before defining the groups.

The first is the add-liquidity-only instruction, written *Alo* and also known as post-only. An order

²¹At available timestamps, replayed best-bid/best-ask prices and displayed amounts matched the raw Tardis data feed at least 99.7% of the time.

carrying this instruction is accepted only if it comes to rest in the book, where it adds displayed depth. If the same order would trade on arrival against a quote already resting on the other side, the venue rejects it rather than filling it. The instruction exists for traders who quote continuously on both sides of the book at a narrow spread. A wallet quoting at such a narrow spread cannot know, when it submits an order, whether the midprice will have moved by the time the order reaches the matching engine. Absent the instruction, an order sent to add depth can arrive after such a move and execute against a quote resting at a stale price, so the wallet consumes the liquidity it intended to supply. The instruction eliminates that outcome: the order either adds depth or is rejected. Figure 1 illustrates the mechanics of an add-liquidity-only order.

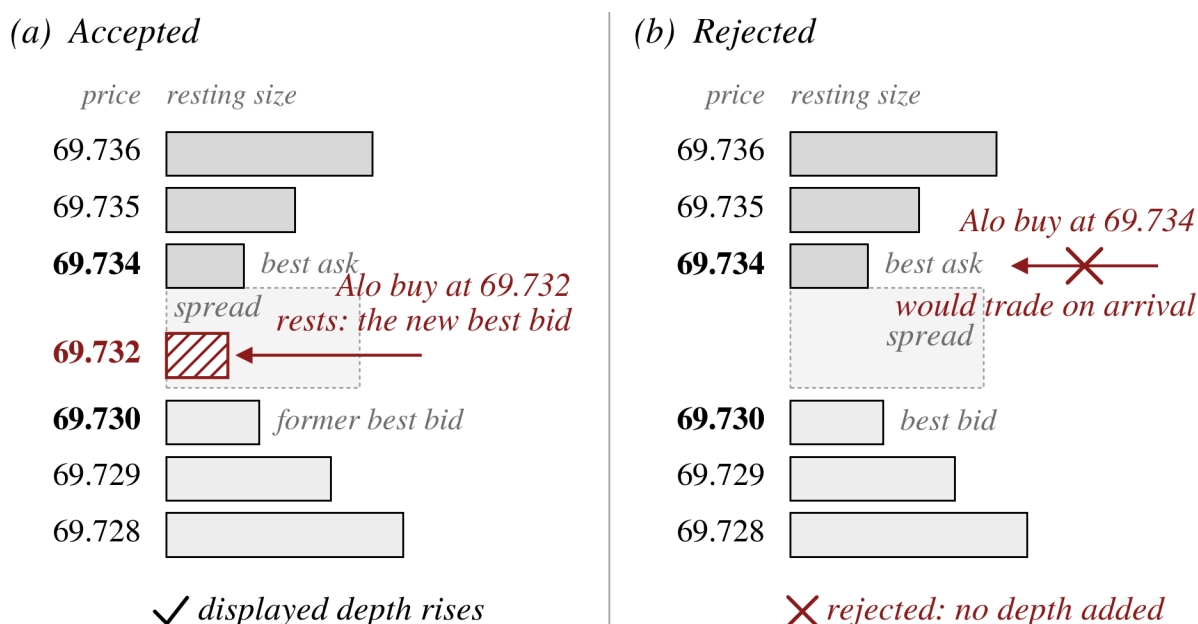


Figure 1: Mechanics of an Add-Liquidity-Only Order

The second is the *FrontendMarket* order type. The Hyperliquid website and application attach this type to a market order entered through them, so the order record distinguishes market orders submitted on the venue's own interface from those submitted through a third-party interface. This route is intended for manual trading: an individual sizes a position on the venue's own interface and submits it directly, with no program between the trader and the matching engine. Figure 2 shows the order-entry panel that produces such an order. A wallet whose orders carry this type has therefore traded, at least in part, manually through this panel, at the pace of human order entry.

Together with trading frequency, these two features supply the criteria that separate traders into groups. Algorithmic market makers are identified first: they are the wallets whose order flow is dominated by *Alo* orders and that quote on both sides of the book almost continuously. Front-end

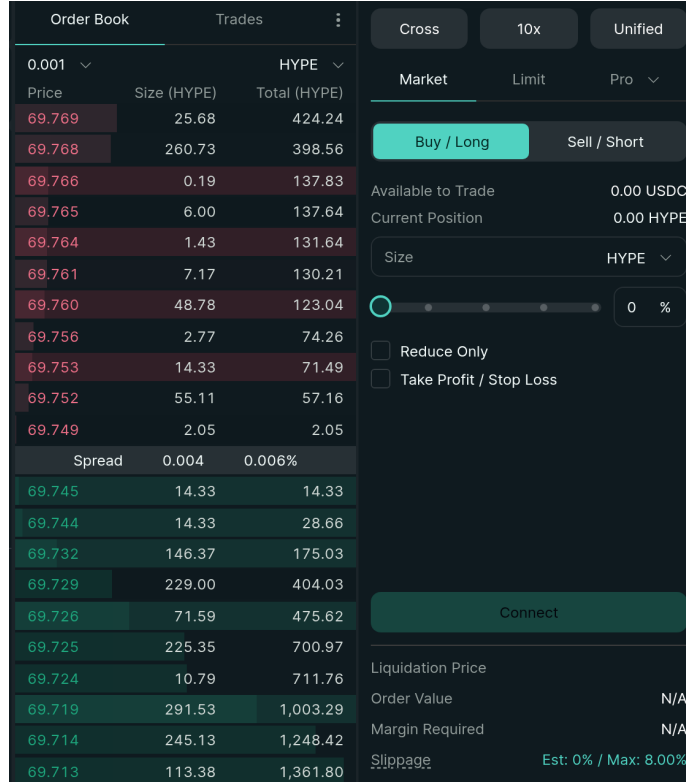


Figure 2: Venue Front-End Order-Entry Panel

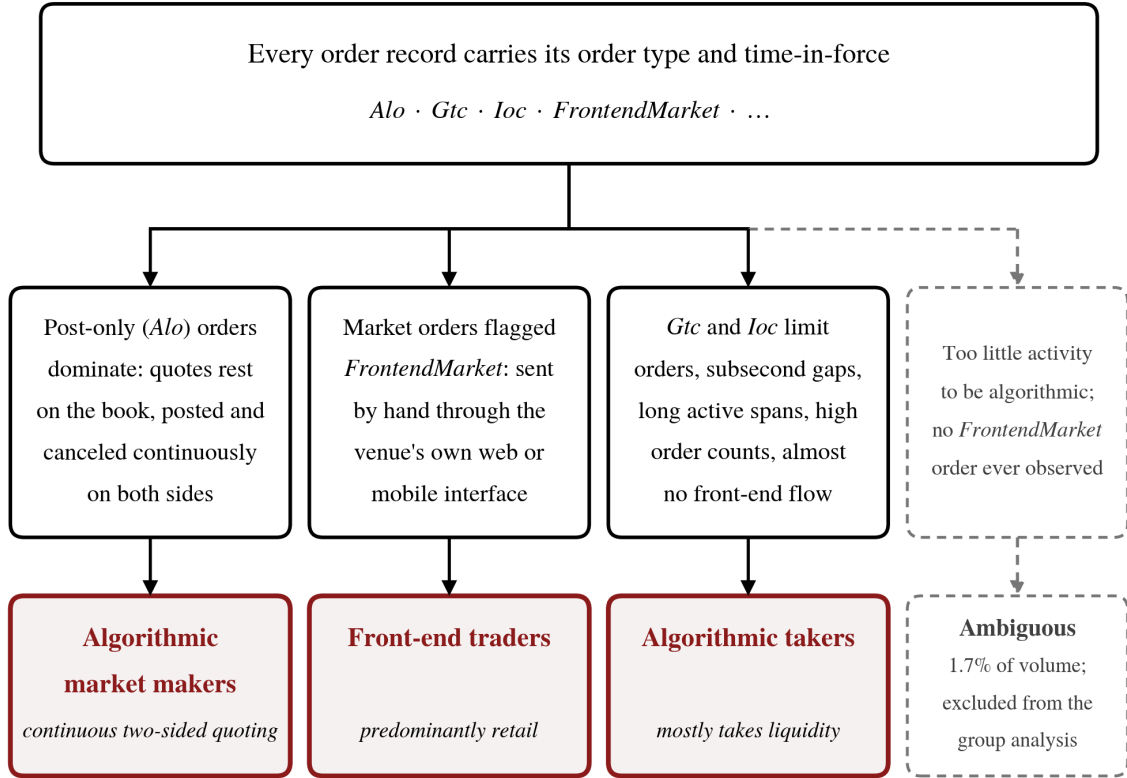
traders are the wallets that submit a substantial share of their orders as *FrontendMarket* orders, and we interpret this group as predominantly retail, because manual entry on the venue’s own interface is the route an individual trader typically uses.²² Algorithmic takers are the automated wallets that mostly take liquidity: they submit *Ioc* (immediate-or-cancel) and *Gtc* (good-till-canceled) orders rather than *FrontendMarket* orders, and their automation is evident in subsecond inter-order gaps, long active-hour spans, high order counts, and persistently low front-end share.²³

The labeling procedure assigns a base label at the (user, coin, date) level, allowing a wallet’s classification to change when its observed trading changes. The two-stage daily labeling procedure uses order types and trading frequency; its exact rules, thresholds, and robustness checks are reported in Section A.1. Figure 3 summarizes the criteria that assign each wallet to a group.

After the first pass, the second pass applies cross-coin persistence rules to the daily base labels. This pass stabilizes labels: it preserves a market-maker or algorithmic-taker classification when a

²²The front-end label is also the conservative default for wallets with too little activity to classify. Wallets that hold the label only through this default and submit no *FrontendMarket* order at any point in the sample form a small residual group, labeled Ambiguous; it accounts for 1.7% of volume, is reported separately in Section A.2, and is excluded from the three-group analysis.

²³This group may be a mixture: interfaces built by third parties also submit *Ioc* orders, so an individual trading through such an interface appears here rather than among front-end traders. The algorithmic rules are deliberately conservative because the absence of the front-end order type is not by itself sufficient to classify a wallet.



*Labels are assigned at the wallet × coin × date level,
so a wallet's label changes when its observed behavior changes.*

Figure 3: Wallet Classification Decision Flow

wallet is temporarily inactive, trades mainly on another coin, or falls just below a per-coin daily threshold.²⁴

²⁴The four Hyperliquid protocol addresses (two HLP vaults and two liquidator engines) are not assigned a behavioral label; the pair-level accounting in Section 6 reports them separately, pooled as *Vault*.

Table 1: Per-Coin Label Coverage across the Top 10 Coins

Coin	Mean users/day			Volume share (%)			Coin total	
	MM	Ret	Algo	MM	Ret	Algo	Active users	Total volume
BTC	387	8,253	338	51.6	42.8	3.7	515,527	\$1.43T
ETH	343	5,008	206	53.8	40.4	4.7	333,981	\$883.69B
SOL	223	3,421	154	54.1	39.6	3.7	303,866	\$267.89B
HYPE	231	4,272	1,131	46.5	43.4	8.5	263,966	\$205.01B
XRP	162	1,192	63	57.9	37.2	2.4	121,256	\$60.84B
FARTCOIN	185	1,145	64	53.0	41.5	3.5	98,530	\$37.19B
DOGE	149	737	51	60.1	35.2	2.9	82,046	\$21.93B
ENA	131	676	34	54.2	40.3	3.6	62,281	\$17.74B
SUI	147	835	51	55.7	37.6	4.9	84,716	\$19.60B
LINK	99	584	33	56.8	37.9	3.2	53,418	\$8.89B
Total							729,540	\$2.95T

Notes. March 27, 2025 to June 30, 2026, October 10 included. “Mean users/day” counts average daily active users by group; “Ret” denotes front-end traders. “Volume share” is the group’s percentage share of the coin’s grouped-trade volume; displayed shares need not sum to 100% because the Ambiguous group is omitted. “Active users” counts distinct wallets active on the coin; the total counts distinct wallets across the ten coins, so it is smaller than the sum of the coin rows. The four Hyperliquid protocol addresses are excluded. Coin total volume is single-sided notional.

Table 1 reports the resulting composition by coin, and the two large groups differ sharply in how many wallets produce their volume. Algorithmic market makers are few, hundreds of daily active wallets on the major coins, yet they account for 46.5% to 60.1% of each coin’s volume. Front-end traders are numerous, thousands of daily active wallets on the major coins, and they account for 35.2% to 43.4% of volume. Algorithmic takers account for a small share of volume on every coin; their largest share is 8.5%, on HYPE.

5.2 Trading Cost Decomposition

Having classified every wallet into a behavioral group, we now measure what each trade costs and what it earns.

The unit of observation is a grouped trade. A marketable order can execute against several resting orders at successively worse prices, with the arriving order the taker in each of those matches and each resting order the maker. The data record each match as a separate fill, so a single order appears in the record as several fills. Those fills reflect one decision by one trader at one moment, and their division into separate fills follows from the state of the book rather than from any choice by the trader. The economic unit is therefore the execution of the order rather than the individual fill. A fill-level design that benchmarks each fill against its own contemporaneous midprice can

understate the execution cost of a marketable order that walks the book, because later benchmarks may already reflect quote revisions induced by the same order. We instead treat all fills generated by the same order at the same timestamp as one execution and benchmark its volume-weighted average price against the strict-prior midprice, retaining the order’s full price concession for the subsequent decomposition into half-spread and slippage.

Let q_i denote trade size, $\bar{p}_i$ the volume-weighted average price (VWAP), and $s_i = +1$ for buys and $s_i = -1$ for sells, signed from the perspective of the leg being measured. Recall from Section 4 that m_t is the strict-prior midprice, read from the replayed book. We write m_{t+h} for the midprice h minutes after the trade.

To measure the outcome of a trade, we study three related objects: trade P&L, price impact, and execution cost. Trade P&L is the trader’s ex-post profit on a trade, signed by direction and scaled by trade size.²⁵ In absolute and relative terms, it is:

$$M_{i,h} = s_i (m_{t+h} - \bar{p}_i) q_i, \quad M_{i,h}^{\text{rel}} = s_i \cdot \frac{m_{t+h} - \bar{p}_i}{m_t}. \quad (1)$$

A positive value means that the trader bought before the midprice rose or sold before the midprice fell. We decompose trade P&L into a price-impact component, which measures the post-trade midprice change, and an execution-cost component, which captures the price concession paid at execution:

$$T_{i,h} = s_i (m_{t+h} - m_t) q_i, \quad \text{EC}_i = s_i (m_t - \bar{p}_i) q_i. \quad (2)$$

We define the relative counterparts of these components analogously, so that the decomposition reads:

$$M_{i,h} = T_{i,h} + \text{EC}_i, \quad M_{i,h}^{\text{rel}} = T_{i,h}^{\text{rel}} + \text{EC}_i^{\text{rel}}. \quad (3)$$

Figure 4 traces the decomposition on the midprice path of a single trade.

For the taker-only execution-cost analysis, we work with the positive cost $\text{EC}_i^+ = -\text{EC}_i$ and decompose it into half-spread and slippage. The *half-spread* is the cost of reaching the first execution price, paid in moving from the prevailing midprice to the best quote resting on the opposite side of the book. The *slippage* is the additional cost from depth consumed beyond that first execution price within the grouped trade, and it increases with the distance the order walks into the book to complete its size. Let $p_{0,i}$ denote the best execution price within grouped trade i : the lowest fill

²⁵ $M_{i,h}$ follows the trader-side P&L convention of (Barber et al., 2009): positive means the trader profited. In industry parlance, this is the (taker) *markout*. In academic microstructure, the closely related *realized spread* measures the same quantity from the liquidity supplier’s perspective and therefore carries the opposite sign under our convention. We use “trade P&L” for consistency with the aggregate P&L accounting in Section 6.

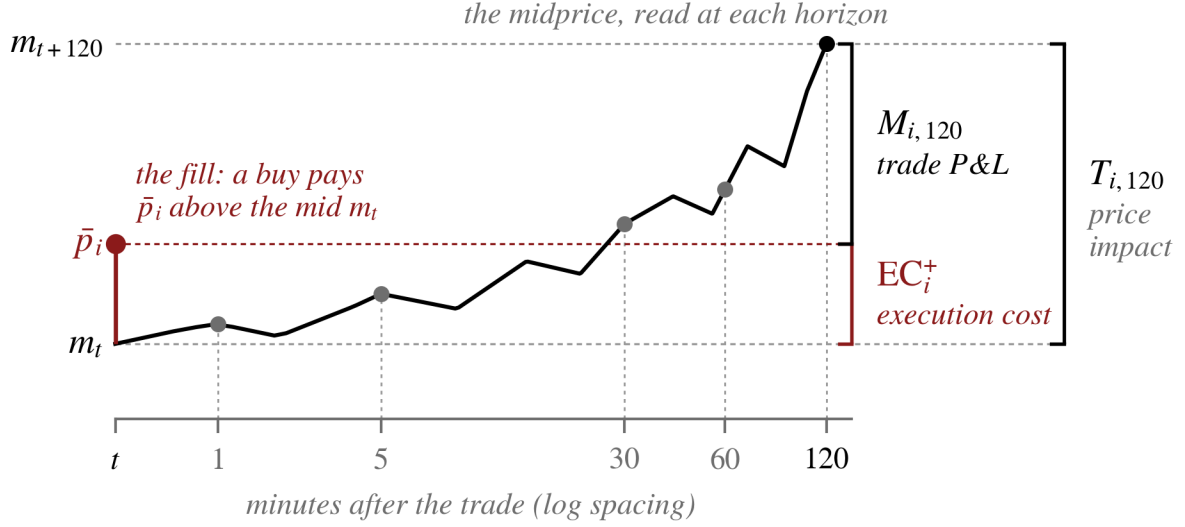


Figure 4: Trade P&L, Price Impact, and Execution Cost on the Midprice Path. The path is the midprice after a taker buy filled at $\bar{p}_i$, above the prevailing mid m_t , with the mid read at each reporting horizon. At the headline horizon the labeled gaps give the decomposition in Equation (3): trade P&L is the midprice move net of the execution cost paid at the fill. The drawn path ends above $\bar{p}_i$, so the depicted buy is profitable; the identity holds for either sign.

price for a taker buy and the highest fill price for a taker sell. We define

$$H_i = s_i(p_{0,i} - m_t)q_i, \quad S_i = s_i(\bar{p}_i - p_{0,i})q_i, \quad EC_i^+ = H_i + S_i = -EC_i. \quad (4)$$

If the event sequence were fully observed, $p_{0,i}$ would coincide with the best opposite quote available when the taker order arrives, so H_i would equal the prevailing book half-spread. Because events that share a timestamp cannot be sequenced, earlier same-timestamp trades may have consumed displayed depth before trade i ; H_i therefore also reflects same-timestamp executions and cancellations that worsen the first available execution price.

Price paths are measured at horizons of 1, 5, 30, 60, and 120 minutes after the trade, and 120 minutes is the headline horizon throughout. Where one of these objects is reported for one side of the trade, or for one category of trades rather than all of them, a superscript names the subset.

5.3 Trade-Purpose Buckets

Each grouped trade is assigned a trade-purpose label using the trader’s signed pre-trade inventory and the closed P&L realized on the fill. *Opening* (Open) trades initiate or increase a position. *Gain-realizing close* (GRC) trades reduce a position and realize positive closed P&L, while *loss-realizing close* (LRC) trades reduce a position and realize negative closed P&L. We classify the two types of forced position reductions separately. *Liquidation* (Liq) trades are protocol-initiated reductions triggered when a trader’s account equity falls below the required margin. *Auto-deleveraging* (ADL) trades are protocol-initiated reductions of profitable positions on the opposite side of the market that help absorb losses not covered by the liquidated trader’s collateral.

Openings, gain-realizing closes, and loss-realizing closes are the *voluntary* trades; liquidations and auto-deleveraging trades are the *forced* trades. Every grouped trade that carries a trade-purpose label falls in exactly one of the two categories, and the analysis reports them separately throughout.²⁶

6 Empirical Results

We first present dollar P&L by group, following the trade-level accounting of (Barber et al., 2009). Table 2 reports 120-minute trade-P&L transfers for each taker–maker pair, while Table 3 decomposes group P&L into voluntary taker and maker trades, forced liquidations, ADL, and fees.²⁷ We then analyze each channel in turn. Section 6.2 studies execution cost. Section 6.3 studies post-trade price impact and flow dynamics. Section 6.4 studies liquidations. Section 6.5 studies fees. Funding, which accrues after the trade rather than at it, is analyzed in Section A.6 in the appendix.

6.1 Dollar Profits and Losses by Group: Front-End Traders Finance the Venue

Front-end traders finance the venue. As takers they lose against every maker group: the largest loss, \$235.37M, is paid on algorithmic-market-maker quotes, in the cell that carries 65.74% of platform notional (Table 2). As makers they lose as well, to both algorithmic groups. Their one favorable cell is the protocol vault, \$111.52M received through ADL, and even there liquidations take back more (\$168.26M), so the vault interaction remains negative on net. The mirror image is the algorithmic market makers: they profit as takers against all three groups, while as makers they earn only from

²⁶A small share of voluntary fills cross the wallet’s position sign within a single trade (e.g., going from +5 to −3). These *reversal* fills combine a close component and an open component and carry no clean trade-purpose label; we drop them from the trade-purpose tables. They account for about 1.6% of voluntary taker trades and 3.1% of voluntary taker notional across the ten coins.

²⁷Tables 20 and 21 in Section A.2 repeat both tables with the Ambiguous group included.

Table 2: Cumulative Dollar Profits and Losses from Trade by (Taker, Maker) Label Pair

Panel A: Trade P&L in \$ millions, bps in parentheses.

Taker group	Maker group			
	MM	Frontend	Algo	Vault
MM	33.09 (+1.11)	45.06 (+1.80)	11.21 (+2.42)	38.88
Frontend	−235.37 (−1.21)	−83.18 (−8.51)	−22.64 (−8.71)	111.52
Algo	24.42 (+1.58)	1.93 (+1.32)	−0.91 (−2.48)	13.57
Vault	48.75	168.26	3.75	−0.54

Panel B: Pair volume in \$ billions, share of platform notional in parentheses.

Taker group	Maker group			
	MM	Frontend	Algo	Vault
MM	298.71 (10.13%)	250.19 (8.48%)	46.31 (1.57%)	2.63 (0.09%)
Frontend	1,938.56 (65.74%)	97.69 (3.31%)	25.99 (0.88%)	7.45 (0.25%)
Algo	154.71 (5.25%)	14.68 (0.50%)	3.67 (0.12%)	0.93 (0.03%)
Vault	5.27 (0.18%)	2.43 (0.08%)	0.20 (0.01%)	0.09 (0.00%)

Notes. March 27, 2025 to June 30, 2026, October 10 included. Panel A reports taker-side M_{120} ; Panel B reports pair volume and its share of total platform notional (2.95T). Trades in which either counterparty is an Ambiguous wallet are omitted, so volume shares need not sum to 100% and the P&L entries need not sum to zero. Vault pools the four protocol addresses; its bps are omitted because most vault volume occurs on October 10.

front-end takers and are adversely selected by algorithmic flow. Algorithmic takers earn against market-maker and front-end quotes and lose only among themselves.

Figure 5 shows that front-end traders’ trade P&L declines almost monotonically and that the loss is concentrated in taker trades. Algorithmic market makers and algorithmic takers accumulate gains more steadily.

Table 4 shows that front-end traders are the only group whose voluntary trade P&L declines with the horizon, from −\$68.3M at five minutes to −\$213.2M at two hours, while both algorithmic groups’ totals rise. The split by role locates the decline on the aggressive side: the front-end taker leg falls to −\$230.5M at two hours while the maker leg turns positive. Because execution cost is fixed at trade time, the widening taker loss is post-trade: the initially favorable price impact of front-end trades declines as the horizon lengthens. We study this channel in Section 6.3.

6.2 Execution Cost

Front-end traders pay \$501.25M of execution cost at the fill, against \$42.59M for algorithmic market makers and \$29.80M for algorithmic takers (Table 3). Per dollar of taker volume, that is

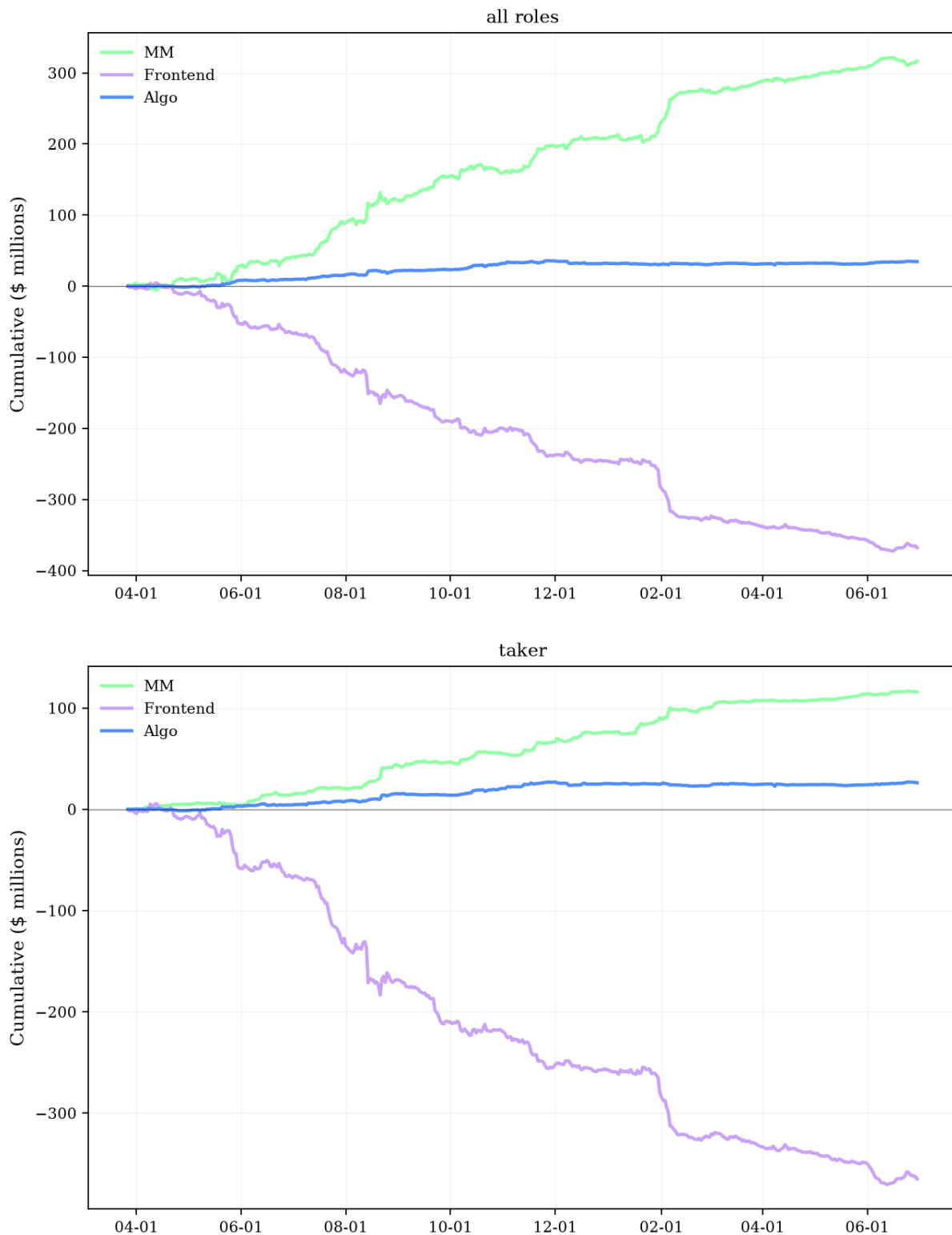


Figure 5: Cumulative dollar profit (and loss) from trade by label, ten-coin sample, March 27, 2025–June 30, 2026, October 10 excluded. Panel A (top): cumulative M_{120} in \$ millions. Panel B (bottom): taker only.

Table 3: Decomposition of Cumulative Dollar Profits and Losses from Trade by Label

Component	MM	Frontend	Algo
T_{120}^{taker}	+148.75	+299.07	+56.86
EC^{taker}	-42.59	-501.25	-29.80
M_{120}^{taker}	+106.16	-202.18	+27.05
M_{120}^{maker}	+190.10	+47.85	+12.76
M_{120}^{Liq}	-79.91	-343.66	-5.28
M_{120}^{ADL}	+41.67	+117.58	+15.89
Total M_{120}	+258.02	-380.42	+50.42
F_{120}	+3.95	-3.82	+0.61
Fees	-50.43	-725.37	-54.90
Net	+211.54	-1,109.61	-3.87

Notes. March 27, 2025 to June 30, 2026, October 10 included. Dollar quantities in millions. T_{120}^{taker} is taker price impact at 120 minutes and EC^{taker} is execution cost, so $M_{120}^{\text{taker}} = T_{120}^{\text{taker}} + EC^{\text{taker}}$. The maker, liquidation, and ADL rows are mutually exclusive; Total M_{120} is the sum over all trades. In each row, F_{120} is the group's 120-minute funding attribution, defined formally in Section A.6. Net equals Total M_{120} plus F_{120} and fees. Totals exclude Ambiguous wallets.

Table 4: Dollar Profits by Horizon

Panel A: Dollar profits, in millions.

Horizon	MM			Frontend			Algo		
	Taker	Maker	Total	Taker	Maker	Total	Taker	Maker	Total
$h = 5$	+106.8	+39.2	+146.0	-37.9	-30.4	-68.3	+13.8	-1.6	+12.2
$h = 30$	+92.5	+152.1	+244.6	-144.8	-13.5	-158.3	+17.1	+6.7	+23.8
$h = 60$	+104.5	+180.2	+284.8	-168.6	-10.1	-178.7	+20.8	+6.6	+27.4
$h = 120$	+117.6	+204.0	+321.5	-230.5	+17.2	-213.2	+24.6	+8.2	+32.8

Panel B: The same cells in bps of each cell's notional.

Horizon	MM			Frontend			Algo		
	Taker	Maker	Total	Taker	Maker	Total	Taker	Maker	Total
$h = 5$	+1.77	+0.16	+0.48	-0.19	-0.84	-0.29	+0.80	-0.21	+0.49
$h = 30$	+1.54	+0.62	+0.80	-0.72	-0.37	-0.66	+0.99	+0.88	+0.96
$h = 60$	+1.73	+0.74	+0.93	-0.83	-0.28	-0.75	+1.21	+0.86	+1.10
$h = 120$	+1.95	+0.83	+1.05	-1.14	+0.47	-0.89	+1.42	+1.07	+1.32

Notes. March 27, 2025 to June 30, 2026, October 10 excluded. Voluntary trades only; liquidation and ADL are excluded. Dollar quantities are in millions; Panel B reports the same cells in bps, calculated using each cell's notional. Total pools the voluntary taker and maker legs.

2.43, 0.70, and 1.70 bps. The mechanism analysis that follows excludes October 10.²⁸

²⁸October 10, 2025 is the day of the marketwide liquidation cascade. On the sample excluding it, the three rates are 2.405, 0.674, and 1.680 bps; Table 5 and every regression in this subsection use that sample. The body rates are the

Table 5: Execution Cost in Bps

Group	Half-spread	Slippage	Execution cost
MM	+0.405	+0.269	+0.674
Frontend	+0.326	+2.080	+2.405
Algo	+0.381	+1.299	+1.680

Notes. March 27, 2025 to June 30, 2026, October 10 excluded. Voluntary taker trades. Positive values are costs. Group taker notional is \$603.16B for MM, \$2.0465T for Frontend, and \$174.26B for Algo.

The half-spread is nearly common across the three groups (Table 5). The gap is in slippage. Front-end traders pay about 7.7 times the algorithmic-market-maker slippage rate per dollar traded, with algorithmic takers in between. The groups differ in the depth their orders consume, not in the quoted spread they cross.

The natural candidate is order size. Figure 6 shows that mean taker slippage rises with the size of the aggressive order (Panel A), and that front-end volume sits farther into the large-order tail of the size distribution than the volume of either algorithmic group (Panel B).

To ask whether larger trades alone account for the premium, we estimate Fama–MacBeth regressions of slippage on trade size, allowing the size relation to vary by group. For the trade-level regressions, we normalize dollar slippage by the trade size valued at the strict-prior midprice:

$$S_i^{\text{rel}} \equiv \frac{S_i}{m_i q_i} = s_i \frac{\bar{p}_i - p_{0,i}}{m_i}. \quad (5)$$

Thus, S_i^{rel} is the cost of execution beyond the grouped trade’s first execution price as a fraction of the trade size valued at the strict-prior midprice; the regression tables report $10^4 S_i^{\text{rel}}$ in basis points, with positive values denoting a cost. Let V_i denote grouped-trade notional and define $v_i = \log(1 + V_i)$. The most flexible specification is

$$S_i^{\text{rel}} = \mu_{c,d,m} + \gamma_{\text{Frontend}} \mathbf{1}_i^{\text{Frontend}} + \gamma_{\text{Algo}} \mathbf{1}_i^{\text{Algo}} + \beta_1 v_i + \beta_2 v_i^2 + \sum_{g \in \{\text{Frontend}, \text{Algo}\}} (\theta_g \mathbf{1}_i^g v_i + \kappa_g \mathbf{1}_i^g v_i^2) + \varepsilon_i. \quad (6)$$

Algorithmic market makers are the omitted group; $\mu_{c,d,m}$ denotes coin-day-minute fixed effects. Table 6 starts from the unconditional gaps, adds common size controls, then group-specific ones. The algorithmic-taker coefficient flips from -0.014 to $+0.052$ once size controls enter: its apparent advantage over algorithmic market makers holds only before size is held fixed.

full-sample convention quoted in the abstract, the introduction, and the conclusion, computed from each group’s dollar execution cost in Table 3 over its taker notional. All rates are reported in the positive-cost convention of Section 5.2.

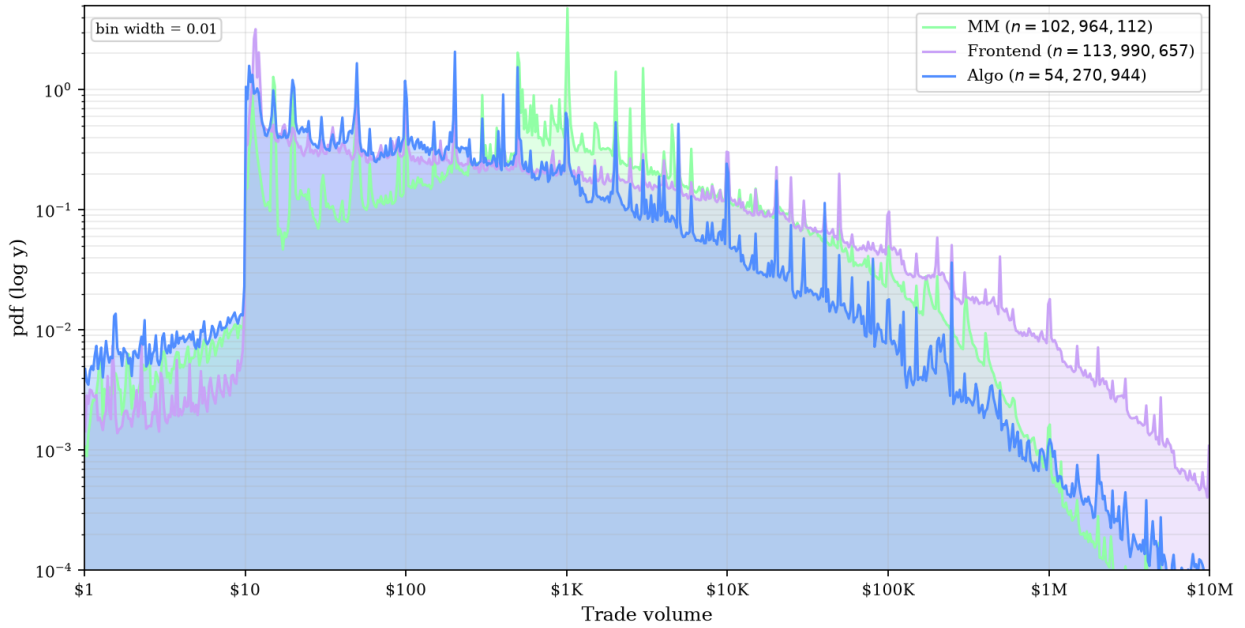
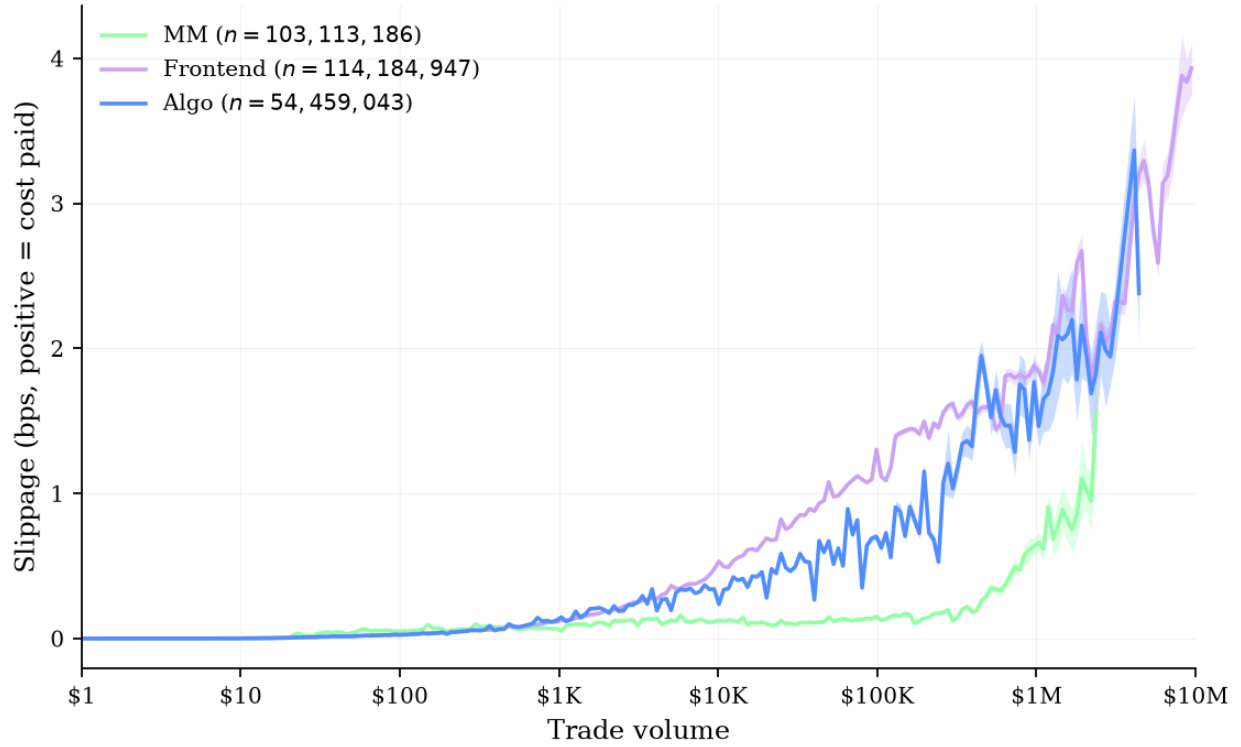


Figure 6: Voluntary taker trades across ten coins, March 27, 2025 to June 30, 2026, with October 10 excluded. Panel A (top): mean taker slippage by log-volume bin (bps, positive = cost), with 95% confidence bands; only group-specific bins with at least 300 trades are shown. Panel B (bottom): volume-weighted distribution of taker trade size.

Table 6: Taker Slippage and Trade Size

	(1) Unconditional	(2) Linear	(3) Quadratic	(4) Group-specific
Frontend	0.196*** (38.26)	0.179*** (47.61)	0.133*** (52.25)	0.350*** (29.18)
Algo	-0.014*** (-11.00)	0.092*** (34.64)	0.052*** (23.16)	0.184*** (18.83)
$\log(1 + V)$		0.076*** (58.41)	-0.145*** (-48.27)	-0.015*** (-8.44)
$\log(1 + V)^2$			0.0168*** (55.75)	0.0032*** (23.38)
Frontend $\times \log(1 + V)$				-0.171*** (-41.57)
Frontend $\times \log(1 + V)^2$				0.0188*** (55.76)
Algo $\times \log(1 + V)$				-0.100*** (-30.16)
Algo $\times \log(1 + V)^2$				0.0114*** (38.24)
Coin-day-minute fixed effects	Yes	Yes	Yes	Yes

Notes. March 27, 2025 to June 30, 2026, October 10 excluded; daily regressions average 494,865 voluntary taker trades. The dependent variable is taker slippage in bps, with positive values denoting a cost. V is taker notional in dollars. Algorithmic market makers are the omitted group. Coefficients are averages of daily estimates; Fama–MacBeth t -statistics are in parentheses. *** denotes significance at the 1% level.

Size accounts for part of the front-end premium, not all of it. The front-end coefficient falls from 0.196 to 0.133 bps under a common quadratic in size, a decline of about one third, and stays significant. The group-specific specification locates the survivor: front-end curvature is +0.0188 against 0.0032 for algorithmic market makers, so the premium sits at the largest front-end trades.

Size explains part of the premium, which leaves a question: why does the same-sized order cost a front-end trader more? The remaining margin is on the liquidity side: whether a trader's order size tracks the depth displayed on the opposite side of the book at that moment. Let $L_{0,i}$ denote the dollar depth immediately before trade i at the best ask for a taker buy and at the best bid for a taker sell, and define $\ell_i = \log(1 + L_{0,i})$. We estimate a panel regression of log trade size on log opposite-side first-level depth, within coin-day-minute:

$$v_i = \mu_{c,d,m} + \gamma_{\text{Frontend}} \mathbf{1}_i^{\text{Frontend}} + \gamma_{\text{Algo}} \mathbf{1}_i^{\text{Algo}} + \beta \ell_i + \sum_{g \in \{\text{Frontend}, \text{Algo}\}} \eta_g \mathbf{1}_i^g \ell_i + \varepsilon_i, \quad (7)$$

where $v_i = \log(1 + V_i)$ is log taker notional, $\mu_{c,d,m}$ denotes coin-day-minute fixed effects, and

algorithmic market makers are the omitted trader group. The coefficient β is the algorithmic-market-maker trade-size elasticity with respect to opposite-side first-level depth, and the interaction terms measure how the front-end and algorithmic-taker elasticities differ from it. Table 7 reports the estimates.

Table 7: Trade-Size Elasticities with Respect to Opposite-Side First-Level Depth

	Coefficient	<i>t</i> -statistic
Frontend	1.083***	33.38
Algo	0.239***	7.08
$\log(1 + L_0)$	0.186***	51.90
Frontend $\times \log(1 + L_0)$	-0.110***	-21.19
Algo $\times \log(1 + L_0)$	-0.184***	-34.29
Coin-day-minute fixed effects	Yes	

Notes. March 27, 2025 to June 30, 2026, October 10 excluded; daily regressions average 494,865 voluntary taker trades. The dependent variable is $\log(1 + V)$, where V is taker notional in dollars. L_0 is opposite-side first-level depth in dollars immediately before the trade. Coefficients are elasticities: a unit change in $\log(1 + L_0)$ maps to a proportional change in trade size. Algorithmic market makers are the omitted group. Coefficients are averages of daily estimates. *** denotes significance at the 1% level.

Algorithmic market makers cut order size when the opposite side is thin; front-end traders barely do. Their largest orders are consistent with size chosen without regard to the depth available: large trades arriving when displayed liquidity is thin. The algorithmic-taker elasticity is near zero, which that group’s much smaller per-trade size makes a plausible reading, though the regression does not itself identify it (Table 28). The crowd-controlled specification in Section A.4 preserves the ordering.

6.3 Post-Trade Price Impact and Flow Dynamics

Recall from Table 4 that the front-end taker loss widens as the horizon lengthens, reaching $-\$230.5\text{M}$ at two hours, while execution cost is fixed at the moment of the trade. Where along the post-trade price path do the three groups separate? Not at the trade. One minute after execution, volume-weighted taker price impact is nearly common across the three groups (Table 8, Total block). By two hours the front-end path has fallen to $+1.23$ bps, while both algorithmic paths end above their one-minute level.

The path between those two endpoints is monotone for two of the three groups. Front-end taker price impact declines at every step of the horizon grid, and algorithmic-taker price impact rises

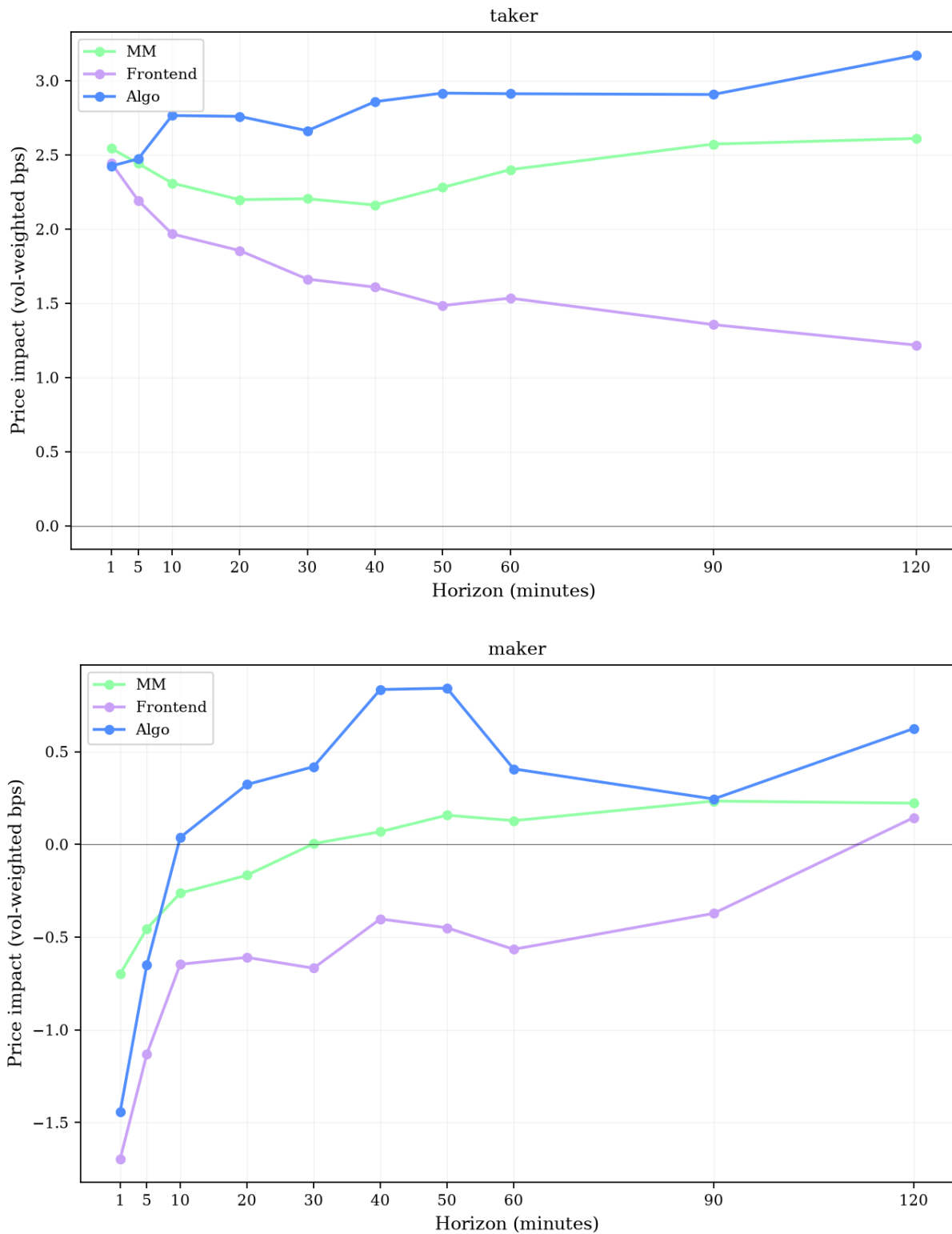


Figure 7: Volume-weighted price impact T_h by horizon and trader group, March 27, 2025 to June 30, 2026. Panel A (taker side, top) and Panel B (maker side, bottom). Positive values indicate favorable post-trade midprice movement. Liquidations, ADL, and October 10 are excluded.

Table 8: Volume-Weighted Taker Price Impact by Trade Purpose

<i>Panel A: Total and openings.</i>						
Horizon	Total			Open		
	MM	Frontend	Algo	MM	Frontend	Algo
$h = 1$	+2.55	+2.44	+2.42	+2.79	+2.20	+2.38
$h = 5$	+2.44	+2.19	+2.47	+2.69	+2.16	+2.62
$h = 30$	+2.21	+1.66	+2.67	+2.61	+1.71	+2.91
$h = 60$	+2.40	+1.55	+2.88	+3.32	+1.70	+3.12
$h = 120$	+2.62	+1.23	+3.09	+3.95	+1.64	+3.57
Volume (\$B)	602.4	2,023.9	172.7	280.0	987.9	85.4

<i>Panel B: Gain-realizing and loss-realizing closes.</i>						
Horizon	Gain-realizing close			Loss-realizing close		
	MM	Frontend	Algo	MM	Frontend	Algo
$h = 1$	+2.29	+2.82	+2.22	+2.28	+2.61	+2.59
$h = 5$	+2.49	+2.70	+2.43	+1.93	+1.86	+2.10
$h = 30$	+2.31	+2.29	+3.84	+1.40	+1.02	+0.95
$h = 60$	+2.54	+2.40	+4.35	+0.72	+0.49	+1.25
$h = 120$	+2.21	+2.37	+4.37	+0.63	−0.71	+1.47
Volume (\$B)	131.2	467.4	41.6	159.4	501.0	37.0

Notes. March 27, 2025 to June 30, 2026, October 10 excluded. Voluntary taker trades. Entries are in bps of taker notional; positive values indicate favorable post-trade midprice movement. Volume is in \$ billions.

at every step. Algorithmic market makers instead dip to +2.21 bps at thirty minutes and recover, ending slightly above their one-minute level. Figure 7 plots the same profiles in bps of notional.²⁹

Which trades carry the divergence? We split voluntary taker trades by trade purpose: openings, gain-realizing closes, and loss-realizing closes.³⁰ Trade purpose is assigned at execution using the trader’s pre-trade position and the realized P&L on the closing component.³¹ Table 8 reports volume-weighted taker price impact for each trader group, trade purpose, and horizon.

The three trade purposes separate the groups in different ways, and openings are the purpose on which the front-end path moves apart from both algorithmic paths. Loss-realizing closes decay sharply for every group. The front-end path is the only one that crosses zero within two hours, at −0.71 bps. Gain-realizing closes hold their level for algorithmic market makers and front-end

²⁹The right-hand panel reports the maker-side counterpart of the same object. The analysis that follows stays on the taker side, where Table 4 located the widening front-end loss.

³⁰Reversal fills, which cross the wallet’s position sign within a single trade, are dropped from trade-purpose tables (see Section 5.3).

³¹Following the disposition-effect literature’s gains-versus-losses realization distinction (Shefrin and Statman, 1985; Odean, 1998), we classify closes as gain-realizing or loss-realizing relative to the wallet’s position reference point. We use this distinction only to classify trade purpose; we do not test asymmetric realization propensities here.

traders, and rise for algorithmic takers. Openings are the disagreement: there the front-end path declines while both algorithmic paths rise.

How do group flows covary with recent returns, in aggregate and within each trade purpose? And whose lagged order-flow imbalances are followed by offsetting flow from the other groups? We answer both questions with reduced-form panel regressions of group flows on recent returns and on other groups' lagged flows.

6.3.1 Reduced-Form Flow Regressions

Within these flow regressions, t indexes consecutive five-minute bins rather than the grouped-trade execution timestamp used elsewhere. Let $\tilde{m}_{c,t}$ denote the replayed-book midprice of coin c at the end of bin t . We define the five-minute log midprice return as

$$r_{c,t} \equiv \log \left(\frac{\tilde{m}_{c,t}}{\tilde{m}_{c,t-1}} \right). \quad (8)$$

The return $r_{c,t}$ is expressed as a decimal fraction, so a log return of 50 basis points is recorded as 0.005. The slot index $s(t)$ identifies the five-minute time slot occupied by bin t within the day, common across coins, and $\eta_{s(t)}^{g,q}$ is the corresponding slot fixed effect. For group $g \in \{\text{MM}, \text{Frontend}, \text{Algo}\}$ and trade purpose $q \in \{\text{Open}, \text{GRC}, \text{LRC}\}$, $V_{\text{buy},c,t}^{g,q}$ and $V_{\text{sell},c,t}^{g,q}$ denote buy and sell taker notional in coin c and bin t , respectively. The denominator $V_{c,t}^{\text{taker}}$ is total taker notional in coin c and bin t . We define signed taker-flow share as

$$\phi_{c,t}^{g,q} = \frac{V_{\text{buy},c,t}^{g,q} - V_{\text{sell},c,t}^{g,q}}{V_{c,t}^{\text{taker}}}. \quad (9)$$

Positive $\phi_{c,t}^{g,q}$ denotes net buying by the cell, while negative values denote net selling, scaled by total taker notional in that coin-bin. For aggregate group flow, we omit q and sum across trade purposes.

Two families of cross-coin panel regressions follow. The first asks whether each flow cell moves with recent returns:

$$\phi_{c,t}^{g,q} = \alpha_c^{g,q} + \eta_{s(t)}^{g,q} + \sum_{\ell=1}^5 \beta_\ell^{g,q} r_{c,t-\ell} + \sum_{\ell=1}^5 \rho_\ell^{g,q} \phi_{c,t-\ell}^{g,q} + \varepsilon_{c,t}^{g,q}. \quad (10)$$

The five return lags trace the short-horizon flow response to recent price movements. The own-flow lags absorb persistence in the same flow cell. A positive β means that the flow moves with recent returns; a negative β means that it trades against them.

The second family asks whether one flow follows, or trades against, another flow:

$$\phi_{c,t}^{g,q} = \alpha_c^{g,q} + \eta_{s(t)}^{g,q} + \theta^{g,q} r_{c,t-1} + \sum_{g',q'} \sum_{\ell=1}^5 \lambda_{\ell}^{g,q \leftarrow g',q'} \phi_{c,t-\ell}^{g',q'} + \varepsilon_{c,t}^{g,q}. \quad (11)$$

The left-hand-side variable is the dependent flow cell. The lagged right-hand-side flows are source flows. Their five lags trace lead–lag interactions across groups and trade purposes, and the lagged return controls for the most recent price movement. A positive λ means that the dependent flow moves with the lagged source flow; a negative λ means that it trades against it.³² All coefficients are estimated daily and averaged across days in Fama–MacBeth form, so inference is at the day level.

The regressions deliver three findings.

The first is at the group level: front-end flow is momentum, and algorithmic flow is contrarian. In Table 9, the coefficient on the first return lag is +6.823 for front-end flow and negative for both algorithmic groups. The net buy share of front-end traders rises after the price has risen and falls after it has fallen, and the algorithmic groups take the other side. The association is short-lived: the front-end coefficient is indistinguishable from zero at the second lag and negative at lags three through five.

Table 9: Aggregate Return-to-Flow Regressions

	(1) ϕ_t^{MM}	(2) ϕ_t^{Frontend}	(3) ϕ_t^{Algo}
r_{t-1}	−1.777*** (0.185)	+6.823*** (0.315)	−1.278*** (0.089)
r_{t-2}	−1.636*** (0.164)	−0.041 (0.245)	−0.752*** (0.084)
r_{t-3}	−1.194*** (0.169)	−0.764*** (0.251)	−0.593*** (0.083)
r_{t-4}	−1.026*** (0.170)	−1.114*** (0.247)	−0.281*** (0.074)
r_{t-5}	−0.731*** (0.133)	−0.907*** (0.235)	−0.292*** (0.080)

Notes. March 27, 2025 to June 30, 2026, October 10 excluded. Return-to-flow specification (Equation (10)): whether each group’s signed flow moves with recent returns. Entries are the coefficients on the five return lags. Daily regressions include coin and slot fixed effects. Fama–MacBeth standard errors in parentheses; *** denotes significance at the 1% level.

The second finding is that momentum has a trade-purpose structure. Splitting flow into the

³²In aggregate regressions the source-flow sum runs over the three trader groups; in trade-purpose regressions it runs over the nine trader-group by trade-purpose cells.

nine group-purpose cells (Table 10) relocates it: loss-realizing-close flow follows returns for every group, and most strongly for front-end traders, at +5.926, more than twice their opening coefficient. Gain-realizing-close flow is contrarian for every group. Openings are the only purpose on which the groups disagree: front-end opening flow follows returns, at +2.176, while both algorithmic groups' opening flow is contrarian. Front-end momentum is therefore shared where every group chases returns, in loss realization, and its own where no other group does, at the open.

Table 10: Return-to-Flow Regressions by Trade Purpose

Trade purpose	(1) $\phi_t^{\text{MM},q}$	(2) $\phi_t^{\text{Frontend},q}$	(3) $\phi_t^{\text{Algo},q}$
Open	−2.491*** (0.115)	+2.176*** (0.192)	−0.590*** (0.055)
GRC	−3.923*** (0.103)	−2.654*** (0.113)	−0.936*** (0.044)
LRC	+3.846*** (0.117)	+5.926*** (0.162)	+0.238*** (0.038)

Notes. March 27, 2025 to June 30, 2026, October 10 excluded. Return-to-flow specification (Equation (10)) estimated separately for each trader-group by trade-purpose cell: whether that cell's signed flow moves with recent returns. Entries are the coefficient on r_{t-1} ; the remaining four return lags and the own-flow lags are included as controls. Daily regressions include coin and slot fixed effects. Fama–MacBeth standard errors in parentheses; *** denotes significance at the 1% level.

The third finding is who trades against whom. In the lead–lag structure (Table 11), lagged front-end flow enters both algorithmic groups' equations negatively: the contrarian side positions against the most recent front-end imbalance. Once those lagged flows enter, the algorithmic-market-maker return coefficient attenuates to −0.137 while the front-end coefficient is nearly unchanged.

Table 11: Cross-Group Flow Regressions

	(1) ϕ_t^{MM}	(2) ϕ_t^{Frontend}	(3) ϕ_t^{Algo}
r_{t-1}	-0.137 (0.188)	+ 6.806 *** (0.314)	- 0.999 *** (0.090)
ϕ_{t-1}^{MM}	+ 0.097 *** (0.002)	+0.001 (0.001)	-0.000 (0.001)
$\phi_{t-1}^{\text{Frontend}}$	- 0.031 *** (0.001)	+ 0.087 *** (0.002)	- 0.006 *** (0.000)
ϕ_{t-1}^{Algo}	- 0.032 *** (0.002)	+ 0.013 *** (0.003)	+ 0.153 *** (0.004)

Notes. March 27, 2025 to June 30, 2026, October 10 excluded. Flow-to-flow specification (Equation (11)): whether one group's flow follows, or trades against, another group's lagged flow. Entries are the first flow lag and the lagged return; lags two through five are included as controls. Daily regressions include coin and slot fixed effects. Fama–MacBeth standard errors in parentheses; *** denotes significance at the 1% level.

At the purpose level (Table 12), lagged loss-realizing-close flow is met by opposing opening and gain-realizing-close flow across the board, while within its own cell it persists, with the three own-cell loadings the largest in the panel. Every sign survives conditioning on the lagged flow of every other cell, and the front-end loss-realizing-close coefficient is nearly unchanged at +5.949.

Table 12: Trade-Purpose Flow-to-Flow Regressions

Panel A: return coefficient after all lagged flow controls.

Trade purpose	$\phi_t^{\text{MM},q}$	$\phi_t^{\text{Frontend},q}$	$\phi_t^{\text{Algo},q}$
Open	−1.059***	+2.531***	−0.426***
GRC	−3.049***	−1.997***	−0.778***
LRC	+4.015***	+5.949***	+0.236***

Panel B: lag-one coefficients on loss-realizing-close source flows.

Dependent flow cell	$\phi_{t-1}^{\text{MM,LRC}}$	$\phi_{t-1}^{\text{Frontend,LRC}}$	$\phi_{t-1}^{\text{Algo,LRC}}$
$\phi_t^{\text{MM,Open}}$	−0.030***	−0.027***	−0.017***
$\phi_t^{\text{MM,GRC}}$	−0.022***	−0.012***	−0.021***
$\phi_t^{\text{MM,LRC}}$	+0.157***	+0.004***	+0.002
$\phi_t^{\text{Frontend,Open}}$	−0.003	−0.021***	−0.011**
$\phi_t^{\text{Frontend,GRC}}$	−0.024***	−0.015***	−0.009***
$\phi_t^{\text{Frontend,LRC}}$	+0.019***	+0.096***	+0.020***
$\phi_t^{\text{Algo,Open}}$	−0.003***	−0.004***	−0.009***
$\phi_t^{\text{Algo,GRC}}$	−0.005***	−0.003***	−0.004***
$\phi_t^{\text{Algo,LRC}}$	+0.003***	+0.000	+0.149***

Notes. March 27, 2025 to June 30, 2026, October 10 excluded. Flow-to-flow specification (Equation (11)) estimated over the nine trader-group by trade-purpose cells. Panel A reports the return coefficient once all lagged cell flows are controlled for; Panel B reports the lag-one loadings on loss-realizing-close source flows. Daily regressions include coin and slot fixed effects.

Two associated margins characterize what happens to front-end voluntary trades after execution. The first is openings. Their return-to-flow coefficient is the only one that differs in sign across groups. Their price-impact path is also the one that declines for front-end traders while rising for both algorithmic groups. The second is loss-realizing closes. Their flow follows recent returns for all three groups, most strongly for front-end traders. Their price impact decays for all three groups, and the front-end path is the only one to cross zero within two hours.

6.4 Liquidations

Section 6.3 identified the loss-realizing close as the voluntary exit whose price impact decays for every group. The protocol enforces that same exit whenever account equity falls below maintenance margin. What the forced version costs, and who pays it, is what that result leaves open. The incidence is concentrated on one group. Front-end traders bear 80.1% of the liquidation losses

recorded for the three labeled groups (Table 3). They lose four times as much to forced exits as the two algorithmic groups combined.

Timing separates the groups as sharply as incidence does. Algorithmic-market-maker liquidations concentrate on a single day: October 10, 2025 carries 79.7% of their liquidation volume (Table 27). Front-end liquidations are spread across the sample, and only 9.3% of front-end liquidation volume falls on that day. Excluding it, front-end traders account for 97.6% of three-group liquidation volume. Scaled by volume, the same asymmetry holds: liquidations reach 1.475% of front-end volume against 0.025% of algorithmic-market-maker volume (Table 27).

Incidence and timing say who is liquidated, not what a forced trade looks like. Forced exits differ from voluntary ones in size, in the price paid for immediacy, and in the density of trading around them. Table 13 ranks the four trade purposes on all three dimensions. Within the voluntary purposes the gradient is mild: average notional and half-spread both rise from openings through gain-realizing closes to loss-realizing closes, reaching \$7.7k and 0.92 bps. Liquidations sit far outside that gradient, at \$31.1k and 15.16 bps. That is four to six times the voluntary trade size, and on the order of twenty times the voluntary half-spread. Liquidations also execute in crowded moments: 83.9% of liquidation trades share their millisecond timestamp with another wallet’s taker trade, against 27.4% of openings.

Table 13: Equal-Weight Cross-Coin Trade-Purpose Descriptives

Trade purpose	$\bar{H}$ (bps)	$\bar{V}$ (\$k)	$P(C > 0)$ (%)
Opening	+0.67	5.5	27.4
Gain-realizing close	+0.69	7.0	30.0
Loss-realizing close	+0.92	7.7	32.0
Liquidation	+15.16	31.1	83.9

Notes. March 27, 2025 to June 30, 2026, October 10 included. Ten coins, taker side. Values are computed within coin before equal-weight averaging. $\bar{H}$ is positive half-spread and $\bar{V}$ is average trade notional. $P(C > 0)$ is the probability that a taker trade shares its millisecond timestamp with another wallet’s taker trade.

The descriptive gap mixes two things: the price of forced execution and the composition of the trades that are forced. What does a liquidation cost at the same trade size, in the same coin and minute? For each day, we estimate

$$Y_i = \mu_{c,d,m} + \delta_d \text{Liq}_i + \beta_d \log V_i + \text{order-type controls} + \varepsilon_i, \quad (12)$$

where Y_i is either taker half-spread or slippage in positive-cost bps, Liq_i indicates a liquidation, V_i is trade notional in dollars, and $\mu_{c,d,m}$ denotes coin-day-minute fixed effects.³³ We report the mean

³³The TIF set $\mathcal{T}$ contains *Alo*, *Ioc*, front-end market, TWAP, missing TIF, and other TIF values. *Gtc* is omitted, and

daily δ_d with its Fama–MacBeth standard error.

Table 14: Conditional Liquidation Cost Premium

Outcome (bps)	Liquidation	
	Coefficient	FM SE
Half-spread	+1.99	(0.115)
Slippage	+0.06	(0.010)

Notes. March 27, 2025 to June 30, 2026, October 10 excluded; 260,733,744 taker trades over 448 days with estimable panels. Outcomes are positive costs in basis points of trade notional. The specification is given in Equation (12); coefficients are averages of daily estimates, with Fama–MacBeth standard errors in parentheses.

At a given trade size and within the same coin-minute, the liquidation indicator is associated with +1.99 bps of half-spread and with +0.06 bps of slippage (Table 14). Both coefficients are small beside the 15.16 bps unconditional liquidation half-spread of Table 13. That contrast is consistent with an important role for trade size and for the conditions prevailing in the minutes when liquidations arrive. The two magnitudes are different estimands measured on different samples.³⁴ They are read alongside each other and never differenced.

Execution cost prices the trade at the fill and says nothing about the position afterwards. The post-trade path of a liquidation runs opposite to the voluntary exit it resembles. Table 15 reports cumulative volume-weighted price impact by trade purpose and horizon. Every voluntary purpose stays positive across the window, and the loss-realizing close decays to +0.66 bps at two hours. Outside October 10, the liquidation path is adverse from the first mark and deepens with the horizon, to −28.48 bps at two hours.³⁵ The gap widens with the horizon.

the liquidation market order type is omitted because it is collinear with Liq_i .

³⁴The descriptive comparison includes October 10 and equal-weights coins, while the regression excludes that day and compares trades inside the same coin-day-minute.

³⁵Including October 10, the path opens at +2.84 bps and stands at −96.70 bps at two hours. From thirty minutes onward the crash day deepens an adverse path that is already present without it, while at five minutes the two conventions disagree in sign.

Table 15: Cumulative Volume-Weighted Price Impact by Trade Purpose

Horizon	Voluntary takers			Liquidation	
	Open	GRC	LRC	ex Oct-10	incl Oct-10
$h = 5$	+2.30	+2.53	+2.50	−16.04	+2.84
$h = 30$	+1.95	+2.25	+1.97	−21.03	−79.56
$h = 60$	+2.26	+2.61	+0.69	−24.21	−97.68
$h = 120$	+2.00	+2.54	+0.66	−28.48	−96.70
Notional	\$1,364B	\$644B	\$704B	\$33.5B	\$37.5B

Notes. March 27, 2025 to June 30, 2026; October 10 included except in the liquidation ex Oct-10 column. Ten-coin pooled; entries are in bps of the notional in the last row, which is in \$ billions. Liquidation counts both roles and is shown with and without October 10.

Gross liquidation losses are not the end of the accounting. Auto-deleveraging (ADL) offsets part of these losses without returning them to the wallets that incurred them. It closes profitable opposite-side positions when a liquidated account cannot be cleared against the book, and it is distinct from the liquidator vault’s unwinding of residual positions (Section 3). For front-end traders, auto-deleveraging contributes +\$117.58M against liquidation losses of \$343.66M (Table 3). That contribution accrues at the group level rather than as restitution to the liquidated wallets. Net forced exits are \$226.08M, the second channel of the front-end shortfall reported in Section 1.

6.5 Fee Tiers

Fees are the largest of the four measured channels for front-end traders in Table 3. Front-end traders pay 87.32% of the fees the three groups pay. The schedule is posted, and it applies to every wallet on the same terms. The burden is nonetheless uneven. All three groups pay fees rather than receive them. The largest component of the front-end shortfall is therefore not an offsetting gain to any group the paper measures. Two margins are associated with that incidence: the side of the trade each group takes, and the distribution of each group’s volume across the reconstructed fee tiers. Role and volume are chosen jointly, so the two margins are reported as observed incidence and not as a causal decomposition of the fee bill.

Dollar totals reflect size as much as terms. The next question is what each group pays per dollar it trades. Measured against each group’s total volume in Table 16, front-end traders pay 2.99 bps and algorithmic market makers pay 0.16 bps. Algorithmic market makers trade the largest volume of the three groups and pay the smallest bill, in dollars and per dollar traded alike. The blended rate front-end traders pay is roughly eighteen times theirs. Algorithmic takers pay a blended rate

within a basis point of the front-end rate, on a tenth of the front-end volume.³⁶ The per-dollar burden separates the two algorithmic groups from each other, not algorithmic wallets from front-end wallets.

Hyperliquid’s fee schedule has two dimensions: how much of the venue’s native token a wallet stakes,³⁷ and the wallet’s recent trading activity, measured by weighted trailing 14-day volume.³⁸ The base taker fee is 4.5 bps and the base maker fee is 1.5 bps. The highest-volume makers earn rebates up to 0.3 bps. For each active user-day, we reconstruct the wallet’s tier from platform-wide trading volume. Fee and funding quantities carry the trader’s sign throughout the paper: a negative entry is an amount the trader pays, and a positive entry is an amount the trader receives or earns.

Table 16: Per-Group Realized Fee Rates and Volume Composition by Role

Group	Taker fee (bps)	Maker fee (bps)	Taker vol	Maker vol	Total vol
MM	−1.71	+0.22	\$610B (20%)	\$2.48T (80%)	\$3.09T
Frontend	−3.32	−0.80	\$2.06T (85%)	\$371B (15%)	\$2.43T
Algo	−2.87	−0.55	\$175B (69%)	\$78B (31%)	\$253B

Notes. March 27, 2025 to June 30, 2026, October 10 included. Liquidation and ADL fills excluded. Percentages in parentheses are each role’s share of the group’s total volume. Positive means earned.

What puts a group on one rate rather than another? The first margin is the side of the trade. Front-end traders trade overwhelmingly as the taker: 85% of their volume crosses the spread (Table 16). The maker–taker schedule charges the aggressor for demanding immediacy. Taker share orders the three groups, and the realized rates order them the same way on both sides of the market. Algorithmic market makers hold the one positive fee entry in the table, and they earn it on the side of the market that carries the majority of their volume. The gap is inside the taker side as well. Front-end traders pay nearly twice the taker rate algorithmic market makers pay on the same side of the same book. That leaves the second margin: where each group’s volume sits in the tier schedule.

Second, front-end traders are concentrated in the highest-fee tier: 92.1% of front-end wallets generate 73.5% of front-end volume. Algorithmic-market-maker volume is concentrated at the lowest-fee diamond tier: 279 wallets, or 8.1% of algorithmic market makers, generate 37.6% of algorithmic-market-maker volume. Algorithmic takers fall between the two: 87.7% are in the highest-fee tier and generate 49.4% of algorithmic-taker volume. The full tier-by-group breakdown appears in Table 23 and Section A.3.

³⁶The algorithmic-taker blended rate is 2.17 bps.

³⁷Staking means locking the venue’s native token (HYPE) with the protocol in exchange for fee discounts. Tiers run from no stake at the base to the largest stake at the top, labeled *none* through *diamond*.

³⁸Weighted trailing 14-day volume is $V_{14} = V_{14}^{\text{perp}} + 2 \cdot V_{14}^{\text{spot}}$, with spot volume counted twice against perpetual volume.

7 Conclusion

Perpetual futures never expire, so a trader holds exposure without rolling into a successor contract, and a funding rate ties the contract price to spot continuously. Exchanges and regulators cite these features as the instrument moves onshore. What the market costs its traders, and who those traders are, has been harder to see: the venues where the contract developed sit outside the reporting infrastructure of regulated markets. Hyperliquid is an exception: it matches orders on a public blockchain, so each trade and forced liquidation is recorded with that account's position and realized profit and loss. Accounts are pseudonymous wallets, so that record becomes an accounting of who gains and who loses only once wallets are classified by how they trade. This paper supplies that classification. Under it, front-end traders, whom we interpret as predominantly retail, finance the venue. They pay the execution cost, the fees, and the funding, and they take the losses when the protocol closes their positions.

Forced exits are where cost and incidence concentrate. Per dollar traded they are the costliest way for a position to be closed, paying an unconditional average half-spread of 15.16 bps. Outside October 10, the day of the marketwide sell-off, front-end traders account for almost all liquidation volume among the three groups. Their liquidations run through the whole sample.³⁹ The European Union's markets regulator has said that perpetual contracts offered to retail clients are likely to fall within its product-intervention measures, which impose leverage limits. Such a limit acts on the same margin condition whose breach forces the exit, and on our classification the incidence it would shift is predominantly retail. This is a start: one venue, one classification, one sample window. Even at that scope, the classification and the accounting built on it establish what a forced exit costs per dollar traded and who bears it.

³⁹The half-spread includes October 10; the volume shares exclude it.

References

- Ackerer, D., J. Hugonnier, and U. Jermann. 2025. Perpetual futures pricing. *Mathematical Finance* 1–17. doi: 10.1111/mafi.70018.
- Aoyagi, J. 2020. Liquidity provision by automated market makers. Working paper, available at SSRN.
- Aquilina, M., S. Foley, L. Gambacorta, and W. Krekel. 2024. Decentralised dealers? examining liquidity provision in decentralised exchanges. BIS Working Paper No. 1227.
- Barber, B. M., X. Huang, P. Jorion, T. Odean, and C. Schwarz. 2024. A (sub)penny for your thoughts: Tracking retail investor activity in TAQ. *The Journal of Finance* 79:2403–27. doi: 10.1111/jofi.13334.
- Barber, B. M., Y.-T. Lee, Y.-J. Liu, and T. Odean. 2009. Just how much do individual investors lose by trading? *Review of Financial Studies* 22:609–32. doi: 10.1093/rfs/hhn046.
- Barber, B. M., S. Lin, and T. Odean. 2024. Resolving a paradox: Retail trades positively predict returns but are not profitable. *Journal of Financial and Quantitative Analysis* 59:2547–81. doi: 10.1017/S0022109023000601.
- Barber, B. M., and T. Odean. 2000. Trading is hazardous to your wealth: The common stock investment performance of individual investors. *The Journal of Finance* 55:773–806. doi: 10.1111/0022-1082.00226.
- Barbon, A., and A. Rinaldo. 2026. On the quality of cryptocurrency markets: Centralized versus decentralized exchanges. *Working paper*.
- Battalio, R., S. A. Corwin, and R. Jennings. 2016. Can brokers have it all? on the relation between make–take fees and limit order execution quality. *The Journal of Finance* 71:2193–238. doi: 10.1111/jofi.12422.
- Boehmer, E., C. M. Jones, X. Zhang, and X. Zhang. 2021. Tracking retail investor activity. *The Journal of Finance* 76:2249–305. doi: 10.1111/jofi.13033.
- Bryzgalova, S., A. Pavlova, and T. Sikorskaya. 2023. Retail trading in options and the rise of the big three wholesalers. *The Journal of Finance* 78:3465–514. doi: 10.1111/jofi.13285.
- Campbell, S., N. Hey, C. C. Moallemi, and M. Nutz. 2026. Risk-based auto-deleveraging. ArXiv:2603.15963.

- Cannon, B., and H. Mohrschladt. 2026. Swimming against the current: Contrarian retail trading. *Journal of Financial and Quantitative Analysis* doi: 10.1017/S0022109026102737. Accepted manuscript.
- Capponi, A., and R. Jia. 2025. Liquidity provision on blockchain-based decentralized exchanges. *The Review of Financial Studies* 38:3040–85. doi: 10.1093/rfs/hhaf046.
- Capponi, A., R. Jia, and S. Yu. 2024. Price discovery on decentralized exchanges. *Review of Financial Studies* doi: 10.1093/rfs/hhag002. Advance article.
- Capponi, A., R. Jia, and B. Zhu. 2024. The paradox of just-in-time liquidity in decentralized exchanges: More providers can lead to less liquidity. Working paper, available at SSRN.
- Cartea, Á., F. Drissi, and M. Monga. 2024. Decentralised finance and automated market making: Execution and speculation. *Journal of Economic Dynamics and Control* Forthcoming; arXiv:2307.03499.
- Chapkovski, P., M. Khapko, and M. Zoican. 2026. Trading gamification and investor behavior. *Management Science* 72:32–56. doi: 10.1287/mnsc.2022.02650.
- Chitra, T. 2026. Autodeleveraging: Impossibilities and optimization. ArXiv:2512.01112.
- CoinGecko. 2026a. 2026 q2 crypto industry report. Data as of July 1, 2026.
- . 2026b. State of crypto perpetuals report 2026.
- Cong, L. W., X. Li, K. Tang, and Y. Yang. 2023. Crypto wash trading. *Management Science* 69:6427–54. doi: 10.1287/mnsc.2021.02709.
- Daian, P., S. Goldfeder, T. Kell, Y. Li, X. Zhao, I. Bentov, L. Breidenbach, and A. Juels. 2020. Flash boys 2.0: Frontrunning in decentralized exchanges, miner extractable value, and consensus instability. In *IEEE Symposium on Security and Privacy*, 910–27. doi: 10.1109/SP40000.2020.00040.
- DefiLlama. 2026. Hyperliquid perps fees, revenue, open interest, and perpetual volume. Accessed August 11, 2026.
- Dyhrberg, A. H., A. Shkilko, and I. M. Werner. 2025. The retail execution quality landscape. *Journal of Financial Economics* 168:104051–. doi: 10.1016/j.jfineco.2025.104051.
- Easley, D., M. O’Hara, S. Yang, and Z. Zhang. 2026. Microstructure and market dynamics in crypto markets. *Journal of Financial Markets* 101071. doi: 10.1016/j.finmar.2026.101071. In press.

- Gornall, W., M. Rinaldi, and Y. Xiao. 2025. Perpetual futures and basis risk: Evidence from cryptocurrency. Working paper, available at SSRN.
- Grant, S. M., J. L. Hobson, and R. K. Sinha. 2024. Digital engagement practices in mobile trading: The impact of color and swiping to trade on investor decisions. *Management Science* 70:2003–22. doi: 10.1287/mnsc.2023.00379.
- Griffin, J. M., and A. Shams. 2020. Is Bitcoin really un-Tethered? *The Journal of Finance* 75:1913–64. doi: 10.1111/jofi.12903.
- Hasbrouck, J., T. J. Rivera, and F. Saleh. 2025a. An economic model of a decentralized exchange with concentrated liquidity. *Management Science* 72:3666–83. doi: 10.1287/mnsc.2024.04510.
- . 2025b. The need for fees at a DEX: How increases in fees can increase DEX trading volume. Working paper, available at SSRN.
- He, S., A. Manela, O. Ross, and V. von Wachter. 2025. Fundamentals of perpetual futures. Working paper, available at SSRN.
- Heimer, R. Z., and A. Simsek. 2019. Should retail investors’ leverage be limited? *Journal of Financial Economics* 132:1–21. doi: 10.1016/j.jfineco.2018.10.017.
- Hu, J., and A. L. Zhang. 2026. Endogenous arbitrage networks in the cryptocurrency exchange market. Working paper, previously circulated as “Competition in the Cryptocurrency Exchange Market”.
- Kaniel, R., G. Saar, and S. Titman. 2008. Individual investor trading and stock returns. *The Journal of Finance* 63:273–310. doi: 10.1111/j.1540-6261.2008.01316.x.
- Kelley, E. K., and P. C. Tetlock. 2013. How wise are crowds? insights from retail orders and stock returns. *The Journal of Finance* 68:1229–65. doi: 10.1111/jofi.12028.
- Kogan, S., I. Makarov, M. Niessner, and A. Schoar. 2024. Are cryptos different? evidence from retail trading. *Journal of Financial Economics* 159:103897–. doi: 10.1016/j.jfineco.2024.103897.
- Lehar, A., and C. Parlour. 2025. Decentralized exchange: The Uniswap automated market maker. *The Journal of Finance* 80:321–74. doi: 10.1111/jofi.13405.
- Lehar, A., C. A. Parlour, and M. Zoican. 2024. Fragmentation and optimal liquidity supply on decentralized exchanges. Working paper, available at SSRN and arXiv.
- Li, T., D. Shin, and B. Wang. 2025. Cryptocurrency pump-and-dump schemes. *Journal of Financial and Quantitative Analysis* 60:3622–59. doi: 10.1017/S0022109025000201.

- Linnainmaa, J. T. 2010. Do limit orders alter inferences about investor performance and behavior? *The Journal of Finance* 65:1473–506. doi: 10.1111/j.1540-6261.2010.01576.x.
- Makarov, I., and A. Schoar. 2020. Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics* 135:293–319. doi: 10.1016/j.jfineco.2019.07.001.
- Malinova, K., and A. Park. 2024. Learning from DeFi: Would automated market makers improve equity trading? Working paper, available at SSRN.
- Milionis, J., C. C. Moallemi, T. Roughgarden, and A. L. Zhang. 2024. Automated market making and loss-versus-rebalancing. Working paper, available at arXiv.
- Odean, T. 1998. Are investors reluctant to realize their losses? *The Journal of Finance* 53:1775–98. doi: 10.1111/0022-1082.00072.
- . 1999. Do investors trade too much? *American Economic Review* 89:1279–98. doi: 10.1257/aer.89.5.1279.
- Park, A. 2023. The conceptual flaws of decentralized automated market making. *Management Science* 69:6731–51. doi: 10.1287/mnsc.2021.02802.
- Ruan, Q., and A. Streltsov. 2025. Perpetual futures contracts and cryptocurrency market quality. Working paper, available at SSRN.
- Schwarz, C., B. M. Barber, X. Huang, P. Jorion, and T. Odean. 2025. The “actual retail price” of equity trades. *The Journal of Finance* 80:2507–41. doi: 10.1111/jofi.13467.
- Shefrin, H., and M. Statman. 1985. The disposition to sell winners too early and ride losers too long: Theory and evidence. *The Journal of Finance* 40:777–90. doi: 10.1111/j.1540-6261.1985.tb05002.x.
- Shiller, R. J. 1993. Measuring asset values for cash settlement in derivative markets: Hedonic repeated measures indices and perpetual futures. *Journal of Finance* 48:911–31. doi: 10.1111/j.1540-6261.1993.tb04024.x.
- Subrahmanyam, A., K. Tang, J. Wang, and X. Yang. 2024. Leverage is a double-edged sword. *The Journal of Finance* 79:1579–634. doi: 10.1111/jofi.13316.
- The Block. 2026a. Hip-3 daily open interest and volume by dex. Accessed August 11, 2026.
- . 2026b. Hyperliquid vs exchanges monthly perpetual volumes. Accessed August 11, 2026.

A Appendix: Supplementary Results and Robustness Checks

A.1 Wallet Classification Rules

This appendix specifies the exact classification rules summarized in Section 5.1. The first pass operates on per-coin daily features and assigns the strong cases. The second pass inspects label-transition days and uses cross-coin features to determine whether the previous day’s label should be carried forward.

Feature notation. All inputs are derived from daily user-order summaries: counts of post-only orders Alo , native front-end market orders `FrontendMarket` (abbreviated `fm`), total order IDs `oid`, inter-order gaps in seconds (`min_gap`, `avg_gap`), the active-hour count and active 4-hour block count. Suffix `_t` denotes the current day’s value. A trailing integer $k \in \{2, 5, 7, 14\}$ on a feature denotes a sum or mean over the user’s last k active days within the trailing 14 calendar days. For example, `sum_alo2` sums Alo over today and the previous active day, and `fm_ratio5` is the ratio $\sum \text{FrontendMarket} / \sum \text{oid}$ over the last five active days. `recent14_n` is the number of active days the user has within the trailing 14 calendar days. `consec7_t` and `consec14_t` are the longest run of contiguous active hours-of-day in the trailing 7- and 14-calendar-day windows.

First pass. Each (user, coin, date) is evaluated against the blocks in Table 17 in the order `MM` $\rightarrow$ `Frontend` $\rightarrow$ `Algo` $\rightarrow$ `default`. The first block whose group test fires fixes the label and the remaining blocks are skipped. The `MM` and `Frontend` tests fire when *any* of their listed clauses is satisfied. The `Algo` test is vote-based: with v the number of satisfied clauses among `algo_a`–`algo_e`, strong-`Algo` fires iff $v \geq 3$ or `algo_z`, except on a cold start day (`recent14_n` ≤ 1) where it fires iff `algo_cold` or `algo_z`. Wallets not classified by any block fall through to the default, the front-end label `Frontend`; the two rule tables write this default as base label 1.

Table 17: First-Pass Classification Rules

Group	Code	Clause
MM	mm_a	$\text{sum_alo2} \geq 1,000$ AND $\text{alo_ratio2} \geq 0.7$
	mm_b	$\text{Alo_t} \geq 200$ AND $\text{alo_ratio_t} \geq 0.7$ AND $\text{min_gap_t} < 1$ s AND $\text{fm_ratio_t} < 0.01$ AND $\text{fm_ratio7} < 0.02$
	mm_c	$\text{Alo_t} \geq 2,000$
	mm_d	$\text{recent14_n} \geq 5$ AND $\text{alo_ratio5} \geq 0.8$ AND $\text{avg_min_gap5} \leq 1$ s AND $\text{fm_ratio5} < 0.01$
Frontend	retail_a	$\text{fm_ratio_t} \geq 0.30$
	retail_b	$\text{fm_ratio7} \geq 0.20$
	retail_c	$\text{fm_ratio2} \geq 0.10$ AND $\text{avg_oid2} \leq 50$
	retail_d	$\text{recent14_n} \geq 3$ AND $\text{oid14} < 30$
Algo ^a	algo_a (speed)	$\text{avg_min_gap5} < 0.5$ s OR $\text{avg_gap5} < 300$ s
	algo_b (blocks)	$\text{avg_blocks5} \geq 5$
	algo_c (persistence)	$\text{consec14_t} \geq 12$ OR $\text{active_hours_t} \geq 12$
	algo_d (volume)	$(\text{avg_oid5} \geq 100 \text{ AND } \text{recent14_n} \geq 3)$ OR $\text{sum_oid2} \geq 500$
	algo_e (low front-end)	$\text{fm_ratio5} < 0.02$ AND $\text{oid7} \geq 100$
	algo_z (standalone)	$\text{consec7_t} \geq 20$ AND $\text{fm_ratio7} < 0.05$ AND $\text{oid7} \geq 50$
	algo_cold (cold start)	$(\text{min_gap_t} < 0.5 \text{ s OR } \text{avg_gap_t} < 300 \text{ s})$ AND $\text{oid_t} \geq 200$ AND $\text{fm_ratio_t} < 0.01$
default	—	assign base label 1

Notes. Tested as a cascade in row order: a wallet is labeled by the first block whose group test fires; any wallet that survives all three blocks receives base label 1. The MM and Frontend tests fire if any one clause holds. The Algo test is vote-based on clauses algo_a–algo_e; see footnote.

^a Let v be the number of satisfied clauses among algo_a–algo_e. Strong-Algo fires iff $(v \geq 3)$ OR algo_z when $\text{recent14_n} \geq 2$. On a cold start ($\text{recent14_n} \leq 1$), strong-Algo fires iff algo_cold OR algo_z.

Second pass. The first pass produces a daily label path for each user–coin pair, but a low-activity day can mechanically push an otherwise stable MM or Algo wallet into base label 1. The second pass therefore revisits only label transitions. We first apply two pre-checks. If the strong front-end tests retail_a or retail_b hold on day t , the wallet is assigned Frontend; this prevents a front-end-heavy day from being carried forward as MM or Algo. If there is no change between yesterday’s final label and today’s first-pass label, we keep the first-pass label and make no further adjustment.

For the remaining cases, we identify days on which the first-pass label changes relative to the previous final label. Each such change has an ordered transition, for example MM $\rightarrow$ Frontend, Frontend $\rightarrow$ MM, Algo $\rightarrow$ Frontend, or Frontend $\rightarrow$ Algo. We then test only the stay rule associated with that ordered pair, using the cross-coin features in Table 18. If the corresponding rule fires, the

transition is treated as temporary and the previous active label is carried forward: MM remains MM under the MM-stay rules, and Algo remains Algo under the Algo-stay rule. If the rule does not fire, we accept the first-pass transition.

Table 18: Second-Pass Stay Rules

Stay rule	Transition pair	Clause (any of)
mm_stay_1	{MM, Frontend}	$xc_sum_alo2 \geq 1,000$ AND $xc_alo_ratio2 \geq 0.7$
		$xc_sum_alo5 \geq 1,000$ AND $xc_alo_ratio5 \geq 0.5$
		$xc_alo_t \geq 2,000$
mm_stay_2	{MM, Algo}	$xc_sum_alo2 \geq 1,000$ AND $xc_alo_ratio2 \geq 0.7$
		$xc_alo_t \geq 2,000$
		$xc_sum_alo5 \geq 2,000$ AND $xc_alo_ratio5 \geq 0.5$
algo_stay ^a	{Algo, Frontend}	$v_{xc} \geq 3$ on five cross-coin clauses paralleling first-pass algo_a–algo_e (see footnote)

Notes. All inputs are cross-coin ($xc_$) features aggregated over the ten paper coins on date t . Each rule is tested only at a transition between the listed pair of base labels; here Frontend denotes provisional base label 1. If a rule fires, today’s label is set to the more-active of the two (MM > Algo > Frontend).

^a The five cross-coin clauses are: $xc_avg_min_gap5 < 0.5$ s OR $xc_avg_gap5 < 300$ s (speed); $xc_avg_blocks5 \geq 5$ OR $xc_active_blocks_t \geq 6$ (blocks); $xc_longest_active_run_14d \geq 12$ OR $xc_active_hours_t \geq 12$ (persistence); ($xc_avg_oid5 \geq 100$ AND $xc_recent14_n \geq 3$) OR $xc_sum_oid2 \geq 500$ (volume); $xc_fm_ratio5 < 0.02$ AND $xc_sum_oid7 \geq 100$ (low front-end). Strong-stay fires iff at least three of these hold.

Stability statistics. Labels are assigned at the coin-day level, so a wallet active on several coins can carry several labels on the same date. A wallet’s *dominant label* is the group that accounts for at least 60% of its active days; we assign the wallet to that group only when this threshold is met. Its *stability* is the share of active days carrying that label. Table 19 reports cross-coin label persistence and the active-day distribution from March 27, 2025 to June 30, 2026. The more coins a wallet trades, the more likely it is to hold different labels across coins on a given active day, which is why persistence in Panel A falls as min_coins rises.

Table 19: Cross-Coin Label Stability and Activity

Panel A: persistence of the dominant label.

min_coins	users	% ≥ 100	% ≥ 95	% ≥ 90	% ≥ 80	% ≥ 70	% ≥ 60
≥ 1	729,540	96.13	97.02	97.59	98.47	99.02	99.52
≥ 3	276,041	93.50	95.38	96.45	97.72	98.56	99.27
≥ 5	121,537	90.60	93.91	95.50	97.23	98.29	99.12
≥ 7	52,005	86.13	91.50	93.72	96.10	97.57	98.74
≥ 10	9,381	74.08	85.02	88.71	92.73	95.44	97.54

Panel B: active days by dominant four-group label.

Group	users	mean	p25	median	p75	p90	mean stability (%)
MM	4,062	74.4	4	20	101	242	84.13
Frontend	409,430	18.8	2	6	19	50	99.43
Algo	6,953	99.7	9	31	180	299	77.79
Ambiguous	309,095	9.0	1	3	8	20	99.73

Notes. March 27, 2025 to June 30, 2026, October 10 included. min_coins is the minimum number of coins on which the wallet is active at least one day; threshold columns report the share whose modal label covers at least that fraction of active coin-days. Panel B uses the dominant-label groups and reports the distribution of active days.

The cross-coin feature feed used by the labeling model narrowed on November 25, 2025 and April 16, 2026, so measured persistence can include label changes caused by those input-regime boundaries rather than behavior alone.

A.2 The Ambiguous Group

Recall from Section 5.1 that Ambiguous wallets hold the base label only through the low-activity default and execute no *FrontendMarket* order in the sample. Tables 20 and 21 extend Tables 2 and 3, respectively, to include the Ambiguous group. These wallets approximately break even on trade P&L (total M_{120} of +\$2.4M) but pay \$59.3M in fees, so their after-fee loss consists almost entirely of fees.

Table 20: Cumulative Dollar Profits and Losses from Trade by (Taker, Maker) Label Pair, Including Ambiguous

Maker Taker	MM	Frontend	Algo	Ambiguous	Vault
MM	+33.09	+45.06	+11.21	−2.78	+38.88
Frontend	−235.37	−83.18	−22.64	−6.07	+111.52
Algo	+24.42	+1.93	−0.91	+0.48	+13.57
Ambiguous	−13.06	−3.25	−0.68	−0.21	+15.79
Vault	+48.75	+168.26	+3.75	+4.68	−0.54

Notes. March 27, 2025 to June 30, 2026, October 10 included. Entries report taker-side M_{120} in \$ millions; rows are the taker group and columns the maker group. Vault pools the four protocol addresses. Pair volumes and bps are reported in Table 2.

Table 21: Decomposition of Cumulative Dollar Profits and Losses from Trade, Including Ambiguous

Label	T_{120}^{taker}	EC^{taker}	M_{120}^{taker}	M_{120}^{maker}	M_{120}^{Liq}	M_{120}^{ADL}	Total M_{120}	Funding ₁₂₀	Fees	Net
MM	+148.75	−42.59	+106.16	+190.10	−79.91	+41.67	+258.02	+3.95	−50.43	+211.54
Frontend	+299.07	−501.25	−202.18	+47.85	−343.66	+117.58	−380.42	−3.82	−725.37	−1,109.61
Algo	+56.86	−29.80	+27.05	+12.76	−5.28	+15.89	+50.42	+0.61	−54.90	−3.87
Ambiguous	−1.39	−11.75	−13.14	+8.91	−12.29	+18.94	+2.41	+0.27	−59.30	−56.62

Notes. March 27, 2025 to June 30, 2026, October 10 included. Dollar quantities in millions; columns as in Table 3. Net equals Total M_{120} plus Funding₁₂₀ and fees.

Two observations suggest that Ambiguous wallets are mostly front-end traders reaching the exchange through third-party applications rather than bots. First, their realized fee rate is unusually high: roughly 6 bps of volume, double front-end traders’ 2.99 bps and above the base-tier taker rate of 4.5 bps, the highest fee on the exchange’s posted schedule. In our dataset, fees may include builder fees charged by third-party platforms, which can add up to 0.1% to the perpetual fills they route.⁴⁰ Second, their orders are predominantly *Ioc*. An order submitted programmatically through the exchange API can carry only *Alo*, *Ioc*, or *Gtc*; the *FrontendMarket* value recorded in the

⁴⁰Hyperliquid, “Builder Codes,” *Hyperliquid Docs*.

same field (Section 4) is attached by the venue’s own interface rather than chosen by the submitter. Because the API has no separate market-order instruction, a market order submitted through a third-party front end is executed as *Ioc*, so *FrontendMarket* identifies front-end traders and no one else.⁴¹

Table 22: Order Behavior of Ambiguous and Frontend Wallets

Metric	Group	p5	p25	Median	p75	p95
TIF share (%)	Ambiguous (<i>Ioc</i>)	0.00	37.86	66.67	100.00	100.00
	Frontend (<i>FrontendMarket</i>)	1.82	22.66	50.00	80.00	100.00
Active days	Ambiguous	1	1	3	8	35
	Frontend	1	2	6	19	83
Orders per day	Ambiguous	1.33	2.00	4.00	8.00	25.91
	Frontend	1.33	2.40	4.47	9.50	43.00

Notes. March 27, 2025 to June 30, 2026, October 10 included. TIF share is *Ioc* for Ambiguous and *FrontendMarket* for Frontend, each divided by total orders. Active days are the total number of distinct labeled calendar days per wallet; orders per day are computed per wallet before taking percentiles.

The share distributions contain 306,856 Ambiguous and 408,649 Frontend wallets with recorded orders. The activity rows contain 307,123 Ambiguous and 408,649 Frontend wallets.

Ambiguous wallets resemble front-end wallets more than either algorithmic group. Median active days are 3 and 6, respectively, against 20 for algorithmic market makers and 31 for algorithmic takers (Panel B of Table 19). Median orders per active day are 4.00 and 4.47 (Table 22). Their use of *Ioc* rather than *FrontendMarket* is consistent with retail orders routed through third-party front ends.

⁴¹Hyperliquid, “Exchange Endpoint,” *Hyperliquid Docs*.

A.3 Fee Tier Composition by Group

Recall from Section 6.5 that a wallet’s fee tier has two dimensions: how much HYPE it stakes, and its weighted trailing 14-day volume $V_{14} = V_{14}^{\text{perp}} + 2 \cdot V_{14}^{\text{spot}}$, which counts spot volume twice against perpetual volume. For every active user-day, we reconstruct V_{14} from platform-wide fills and assign the implied volume tier. We allow a $\pm 20\%$ band around adjacent cutoffs to absorb boundary noise since Hyperliquid aggregates volume across master-account sub-wallets that we cannot observe. Labels are estimated by coin and can differ across a wallet’s activity on a given day. We therefore aggregate to the user-day level by assigning each (user, date) the label with the largest cross-coin trade volume that day. Tables 23 and 24 and Figure 8 report the resulting composition.

We call a wallet stable if it has a label on at least 60% of its active days and drop users without a dominant label. The resulting sample contains 3,459 algorithmic market makers, 407,322 front-end wallets, and 6,402 algorithmic takers.

Table 23: User Count and Volume Share by Fee Tier and Group

Fee tier	MM			Frontend			Algo		
	<i>n</i> users	user %	vol. %	<i>n</i> users	user %	vol. %	<i>n</i> users	user %	vol. %
none	2,163	62.53	23.97	375,322	92.14	73.55	5,612	87.66	49.40
wood	105	3.04	0.11	9,751	2.39	2.05	117	1.83	2.01
bronze	151	4.37	0.45	7,313	1.80	4.51	238	3.72	11.54
silver	476	13.76	3.98	3,275	0.80	5.54	88	1.37	5.94
gold	191	5.52	12.13	4,293	1.05	6.66	168	2.62	14.10
platinum	94	2.72	21.81	1,387	0.34	6.37	21	0.33	8.79
diamond	279	8.07	37.55	5,981	1.47	1.34	158	2.47	8.23
total	3,459	100	100	407,322	100	100	6,402	100	100

Notes. March 27, 2025 to June 30, 2026, October 10 included. Ten-coin universe. Stable wallets have a dominant-label ratio of at least 60%. Each wallet is assigned its modal label and modal all-time fee tier. User shares are within group; volume shares are each tier’s share of the group’s total ten-coin trade volume. Ambiguous wallets are excluded.

Table 24: Average Daily Volume by Fee Tier and Group

Fee tier	MM	Frontend	Algo
none	6.38M	271K	98.1K
wood	1.18M	157K	826K
bronze	1.44M	388K	326K
silver	2.00M	820K	379K
gold	15.15M	1.14M	1.60M
platinum	27.06M	3.14M	3.50M
diamond	35.53M	226K	2.29M

Notes. March 27, 2025 to June 30, 2026, October 10 included. Cells are mean ten-coin trade volume in dollars per active user-day. Wallets must have a dominant-label ratio of at least 60% and are assigned to their dominant group; rows use the fee tier observed on that user-day. Ambiguous wallets are excluded.

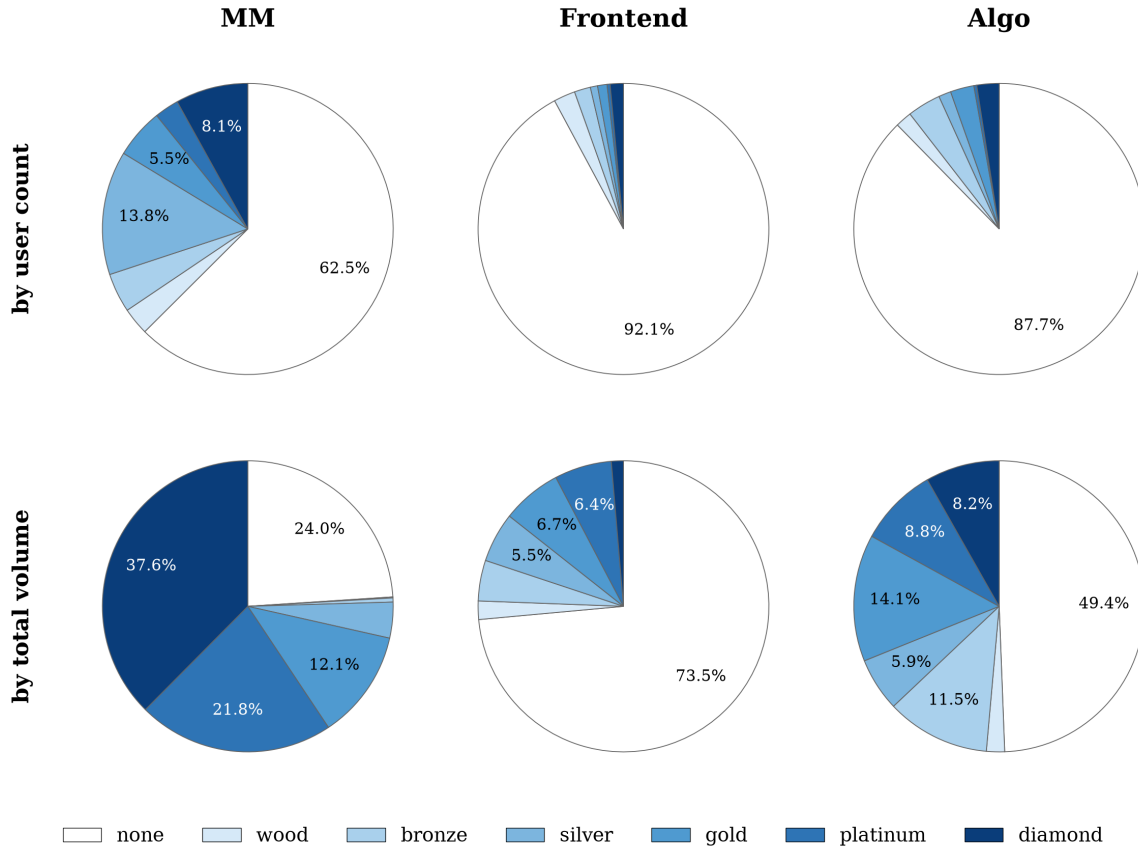


Figure 8: Fee-tier composition by group, March 27, 2025 to June 30, 2026, October 10 included. Top row: user-count shares within group. Bottom row: trade-volume shares within group. Tier colors run from light (*none*) to dark (*diamond*).

A.4 Trade-Size Elasticity with Simultaneous-Crowd Controls

The baseline depth regression in (7) relates each group’s aggressive trade size to same-timestamp available liquidity depth L_0 . A natural concern is that periods of deep liquidity tend to coincide with bursts of taker activity from other participants, so the depth coefficient β may partly reflect contemporaneous co-flow rather than depth-conditioned sizing. We address this with two regressions that condition on the simultaneous same-side taker notional C_i , excluding trade i itself. The first regression augments (7) with $\log(1 + C_i)$ and its group interactions:

$$v_i = \mu_{c,d,m} + \gamma_{\text{Frontend}} \mathbf{1}_i^{\text{Frontend}} + \gamma_{\text{Algo}} \mathbf{1}_i^{\text{Algo}} + \beta \ell_i + \sum_{g \in \{\text{Frontend}, \text{Algo}\}} \eta_g \mathbf{1}_i^g \ell_i + \delta \log(1 + C_i) + \sum_{g \in \{\text{Frontend}, \text{Algo}\}} \psi_g \mathbf{1}_i^g \log(1 + C_i) + \varepsilon_i. \quad (13)$$

The second regression retains the same regressors but redefines the dependent variable as the trade’s size net of the simultaneous crowd, $\log(1 + V_i) - \log(1 + C_i)$. A positive depth coefficient in this regression means a group’s trade is *larger than the contemporaneous crowd* when depth is high; a negative coefficient means it is smaller.

Table 25 shows that the depth-sensitivity ordering survives controls for simultaneous same-side taker activity.

Table 25: Crowd-Controlled Trade-Size Elasticity

	Coefficient	<i>t</i> -statistic
Frontend	1.122***	38.00
Algo	0.229***	7.36
$\log(1 + L_0)$	0.153***	45.04
$\text{Frontend} \times \log(1 + L_0)$	-0.076***	-14.70
$\text{Algo} \times \log(1 + L_0)$	-0.154***	-29.20
$\log(1 + C)$	0.090***	46.16
$\text{Frontend} \times \log(1 + C)$	-0.073***	-32.19
$\text{Algo} \times \log(1 + C)$	-0.031***	-9.44
Coin-day-minute fixed effects	Yes	

Notes. March 27, 2025 to June 30, 2026, October 10 excluded; daily regressions average 494,865 voluntary taker trades. The dependent variable is $\log(1 + V)$, where V is taker notional in dollars. L_0 is opposite-side first-level depth in dollars and C is simultaneous same-side taker notional in dollars, excluding the trade itself. Algorithmic market makers are the omitted group. Coefficients are averages of daily estimates. *** denotes significance at the 1% level.

Controlling for simultaneous activity lowers the algorithmic-market-maker depth elasticity

from 0.186 to 0.153. The implied front-end elasticity remains 0.077, while the algorithmic-taker elasticity remains approximately zero.

Table 26: Relative-Size Regression

	Coefficient	<i>t</i> -statistic
Frontend	0.686***	17.39
Algo	-0.178***	-3.55
$\log(1 + L_0)$	-0.136***	-40.61
$\text{Frontend} \times \log(1 + L_0)$	0.252***	50.48
$\text{Algo} \times \log(1 + L_0)$	0.132***	19.35
Coin-day-minute fixed effects	Yes	

Notes. March 27, 2025 to June 30, 2026, October 10 excluded; daily regressions average 494,865 voluntary taker trades. The dependent variable is $\log(1 + V) - \log(1 + C)$, the trade's size net of the simultaneous same-side crowd. V is taker notional in dollars, L_0 is opposite-side first-level depth in dollars, and C is simultaneous same-side taker notional in dollars, excluding the trade itself. Algorithmic market makers are the omitted group. Coefficients are averages of daily estimates. *** denotes significance at the 1% level.

Table 26 reports the relative-size specification, which sharpens the contrast. Algorithmic-market-maker trade size falls relative to the simultaneous crowd as depth increases, with a slope of -0.136 . The corresponding front-end slope is 0.116, while the algorithmic-taker slope is again approximately zero.

A.5 Liquidation Statistics

Recall from Section 6.4 that front-end traders bear most liquidation losses, while algorithmic-market-maker liquidations concentrate on October 10. Table 27 reports the per-group liquidation frequencies behind those figures. Total volume and trades count both roles and include every trade purpose (voluntary, liquidation, and ADL). Liquidations account for 1.475% of front-end volume and 0.52% of front-end trades, or one trade in 192. For algorithmic market makers they account for only 0.025% of volume.

Table 27: Liquidation Statistics by Group

Group	Liq vol (%)	Total vol	Liq trades (%)	Total trades	Oct-10 liq vol (%)
MM	\$768.6M (0.025%)	\$3,077.10B	2,771 (0.0005%)	563.06M	\$0.61B (79.7%)
Frontend	\$36,024.5M (1.475%)	\$2,442.18B	732,718 (0.52%)	140.88M	\$3.34B (9.3%)
Algo	\$737.7M (0.293%)	\$251.89B	8,620 (0.0076%)	113.46M	\$0.10B (13.3%)
Pooled	\$37,530.8M (0.650%)	\$5,771.16B	744,109 (0.091%)	817.39M	\$4.05B (10.8%)

Notes. March 27, 2025 to June 30, 2026, October 10 included. Both roles. “Liq vol (%)” reports each group’s liquidation volume and, in parentheses, its share of the group’s total volume; “Liq trades (%)” reports the liquidation trade count and, in parentheses, its share of the group’s total trades; “Oct-10 liq vol (%)” reports October 10, 2025 liquidation volume and, in parentheses, its share of the group’s liquidation volume. Total volume and trades include every trade-purpose bucket.

Liquidation per-trade size vs ordinary trades. Table 28 reports the distribution of per-trade notional by trade purpose and group, equal-weight across the ten coins. Three patterns stand out. First, front-end traders have a far longer right tail than the other two groups: at every voluntary purpose their 95th percentile is the largest of the three, from 1.5 times the algorithmic-market-maker value at openings to 13 times the algorithmic-taker value at loss-realizing closes. The contrast with the centre of the distribution is sharpest at openings, where the front-end median trade is \$109 against \$460 for algorithmic market makers while the front-end 95th percentile is the higher of the two. Second, for front-end traders closes are larger than openings, and loss-realizing closes are the largest voluntary bucket: their mean rises from \$7.8k at openings to \$15.3k at gain-realizing closes and \$18.4k at loss-realizing closes. The other two groups show no such pattern; algorithmic-market-maker mean size falls slightly across the three voluntary buckets, from \$3.4k at openings to \$3.1k at loss-realizing closes. Third, liquidations are by far the largest trades for every group. This is most extreme for algorithmic market makers, whose liquidation mean of \$203k is 59 times their opening mean, because 79.7% of algorithmic-market-maker liquidation volume is concentrated on October 10 (Section 6.4); their liquidation median of \$3.6k against that mean shows how few trades carry it. Front-end liquidations are the largest in count and are spread throughout the sample.

Table 28: Notional Quantiles by Trade Purpose

Trade purpose	Group	Mean	p5	p25	p50	p75	p95
Opening	MM	3,438	12	67	460	1,747	12,709
	Frontend	7,834	11	21	109	1,000	19,247
	Algo	1,839	11	20	68	392	3,911
Gain-realizing close	MM	3,400	12	60	411	1,582	12,580
	Frontend	15,294	12	60	465	2,957	41,405
	Algo	2,119	11	24	100	499	5,086
Loss-realizing close	MM	3,092	11	50	373	1,397	10,926
	Frontend	18,405	12	58	472	3,234	49,980
	Algo	1,943	11	20	71	399	3,851
Liquidation	MM	203,366	12	301	3,580	39,165	515,651
	Frontend	35,343	23	239	1,297	8,689	87,618
	Algo	43,808	4	84	842	9,428	103,797

Notes. March 27, 2025 to June 30, 2026, October 10 included. Ten coins, both roles; per-trade notional in dollars. Statistics are computed within coin and then equal-weighted across coins.

A.6 Funding

Funding is the recurring cash flow that ties the perpetual to its reference spot price. Execution cost, fees, and forced exits are realized at the moment of trade. Funding accrues for as long as a position is held. Is funding small or large? The answer depends on which question is asked, and the paper reports two measures. Under the 120-minute attribution that enters the ledger of Table 3, front-end traders pay \$3.82M, a figure already inside their net. Under a position-level exercise that accrues funding through the end of the sample, the same group pays \$284.7M, a figure that sits outside that net. The two are alternative measurements of one channel under different holding assumptions. We report them separately and do not sum them.

Recall the mechanism of Section 3: each hour, every open position accrues a payment equal to its signed size times the oracle price times the settled funding rate for that coin and hour. We reconstruct that payment from the exchange’s settled hourly funding rates and its minute-level oracle prices, both covering the full sample. For each trade i in coin c , executed at time t , define its 120-minute dollar funding attribution as

$$F_{i,120} = -s_i q_i \sum_{H \in (t, t+120m]} f_c(H) P_c^{\text{oracle}}(H), \quad (14)$$

where H runs over the two hourly funding settlements in $(t, t + 120m]$, $f_c(H)$ is coin c ’s settled funding rate at settlement H , and $P_c^{\text{oracle}}(H)$ is its oracle price at that settlement. The quantity $F_{i,120}$ is a 120-minute attribution convention, not the funding accrued over trade i ’s realized holding period. It applies funding to the trade’s signed quantity as if that quantity were held through exactly the two hourly settlements after execution; under the signed-cash-flow convention stated in Section 6.5, positive $F_{i,120}$ denotes a receipt. Tables 3 and 29 report the sum of $F_{i,120}$ within each displayed group under the heading F_{120} . The 120-minute measure applies funding to each trade’s position as if that position were held for exactly two hours. That is an attribution convention. It places funding on the same per-trade footing as the other ledger columns rather than measuring the funding a position accrued. Table 29 reports both measures.⁴²

⁴²Recall the sign convention of Section 6.5: a positive funding figure is received by the trader and a negative figure is paid.

Table 29: Funding by Label and Horizon

Label	120-minute	Held to period end		
		New trades	Initial inventory	Total
MM	+3.95	+72.6	+13.8	+86.4
Frontend	−3.82	−237.9	−46.8	−284.7
Algo	+0.61	+57.4	+31.0	+88.4

Notes. March 27, 2025 to June 30, 2026, October 10 included; dollar funding flows in millions, positive means received. The 120-minute column applies funding to each trade’s position over the two hourly settlements after execution. The held-to-period-end columns accrue funding on positions through June 30, split between inventory already held at the window start and positions opened by trades within the window.

Table 30: Opening Flow and Daily Position Exposure

Group	Open Flow		Net Position		Daily Gross Exposure	
	Long	Short	Mean	Median	Mean	Median
MM	117.70	165.69	−0.409	−0.358	0.825	0.758
Frontend	633.22	373.36	+0.882	+0.694	3.990	3.750
Algo	39.06	47.70	−0.373	−0.211	0.612	0.482

Notes. Opening flow is the total notional of opening trades from March 27, 2025 to June 30, 2026; daily positions are averages of the daily total across all wallets holding a position and cover September 27, 2025 to June 30, 2026, the window over which position snapshots are available. Dollar notional is in billions. Daily positions use the final exchange snapshot.

How large is funding next to the channels that precede it? Under the two-hour convention it is small. In Table 29, front-end traders are the only one of the three groups that pays under this convention. Both algorithmic groups receive. The \$3.82M front-end figure is smaller than the same group’s fee bill by a factor of roughly 190. That smallness is why funding is not one of the paper’s three headline channels. It is nonetheless the funding measure that sits inside the net of Table 3.

Which side of the market does each group hold? Table 30 reports opening flow and daily positions. Front-end traders open long far more than short and carry a positive mean daily net position. Algorithmic market makers and algorithmic takers are net short on average. The medians deliver the same ordering, so the long tilt of the front-end book is a standing feature of the panel rather than the residue of a few extreme days. That direction is consistent with the sign pattern of Table 29, in which the persistently long group pays and the persistently short groups receive.⁴³

⁴³The position columns of Table 30 begin only on September 27, 2025, and therefore cover a shorter window than the funding totals themselves.

What does funding come to if the clock runs to the end of the sample rather than for two hours? The held-to-end exercise changes the holding assumption rather than the data. It accrues funding on reconstructed positions through June 30, 2026. Front-end traders then pay \$284.7M, about 39% of that group's fee bill. Most of it accrues on positions opened inside the window rather than on inventory held when the window opens. Both algorithmic groups receive under this exercise, and the three displayed groups do not net to zero. We do not read those receipts as the counterpart of the front-end cost. The group on the long side also carries the largest book, by mean and median daily gross exposure alike.

Funding therefore reads differently under the two measures. The difference is one of assumed duration rather than of the underlying flow. Under the headline's two-hour attribution it is the smallest column of the ledger. Under the held-to-end exercise it is material next to the channels that precede it.⁴⁴ The two figures answer different questions about one channel, and are reported side by side rather than summed.

⁴⁴The two measures assign wallets to groups by different rules: the 120-minute attribution uses the active user-day classification, and the held-to-end reconstruction assigns each wallet its dominant label over the sample.